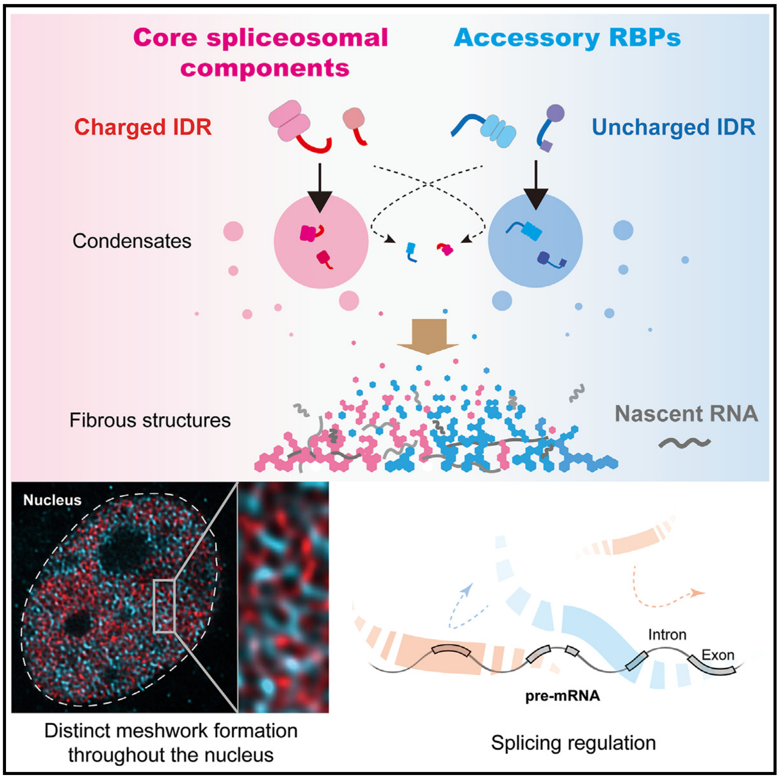
Molecular Cell

Article

Blending and separating dynamics of RNA-binding proteins develop architectural splicing networks spreading throughout the nucleus

Graphical abstract



Authors

Akio Masuda, Takaaki Okamoto, Toshihiko Kawachi, Jun-ichi Takeda, Tomonari Hamaguchi, Kinji Ohno

Correspondence

masuda.akio.h1@f.mail.nagoya-u.ac.jp

In brief

Masuda et al. show that core and accessory RBPs in splicing develop two distinct meshworks throughout the nucleus with charged and uncharged IDRs, respectively. These meshworks mutually exclusively associate with pre-mRNA to conduct splicing. The spatial organization of RBP networks is a functional regulator of RNA splicing.

Highlights

- Core and accessory RBPs constitute two distinct meshworks throughout the nucleus
- The charged/uncharged IDRs of RBPs achieve mutual exclusion in RBP condensation
- RBP binding to nascent RNA induces the formation of fibrous structures
- The RBP meshworks compete for spatial occupancy on pre-mRNA to regulate splicing

Masuda et al., 2024, Molecular Cell *84*, 2949–2965
August 8, 2024
https://doi.org/10.1016/j.molcel.2024.07.001

CellPress

Molecular Cell

CellPress

Article

Blending and separating dynamics of RNA-binding proteins develop architectural splicing networks spreading throughout the nucleus

Akio Masuda,[1,3,4,*] Takaaki Okamoto,[1,3] Toshihiko Kawachi,[1] Jun-ichi Takeda,[1] Tomonari Hamaguchi,[1] and Kinji Ohno[1,2]
[1]Division of Neurogenetics, Center for Neurological Diseases and Cancer, Nagoya University Graduate School of Medicine, Nagoya, Japan
[2]Graduate School of Nutritional Sciences, Nagoya University of Arts and Sciences, Nisshin, Japan
[3]These authors contributed equally
[4]Lead contact
*Correspondence: masuda.akio.h1@f.mail.nagoya-u.ac.jp
https://doi.org/10.1016/j.molcel.2024.07.001

SUMMARY

The eukaryotic nucleus has a highly organized structure. Although the spatiotemporal arrangement of spliceosomes on nascent RNA drives splicing, the nuclear architecture that directly supports this process remains unclear. Here, we show that RNA-binding proteins (RBPs) assembled on RNA form meshworks in human and mouse cells. Core and accessory RBPs in RNA splicing make two distinct meshworks adjacently but distinctly distributed throughout the nucleus. This is achieved by mutual exclusion dynamics between the charged and uncharged intrinsically disordered regions (IDRs) of RBPs. These two types of meshworks compete for spatial occupancy on pre-mRNA to regulate splicing. Furthermore, the optogenetic enhancement of the RBP meshwork causes aberrant splicing, particularly of genes involved in neurodegeneration. Genetic mutations associated with neurodegenerative diseases are often found in the IDRs of RBPs, and cells harboring these mutations exhibit impaired meshwork formation. Our results uncovered the spatial organization of RBP networks to drive RNA splicing.

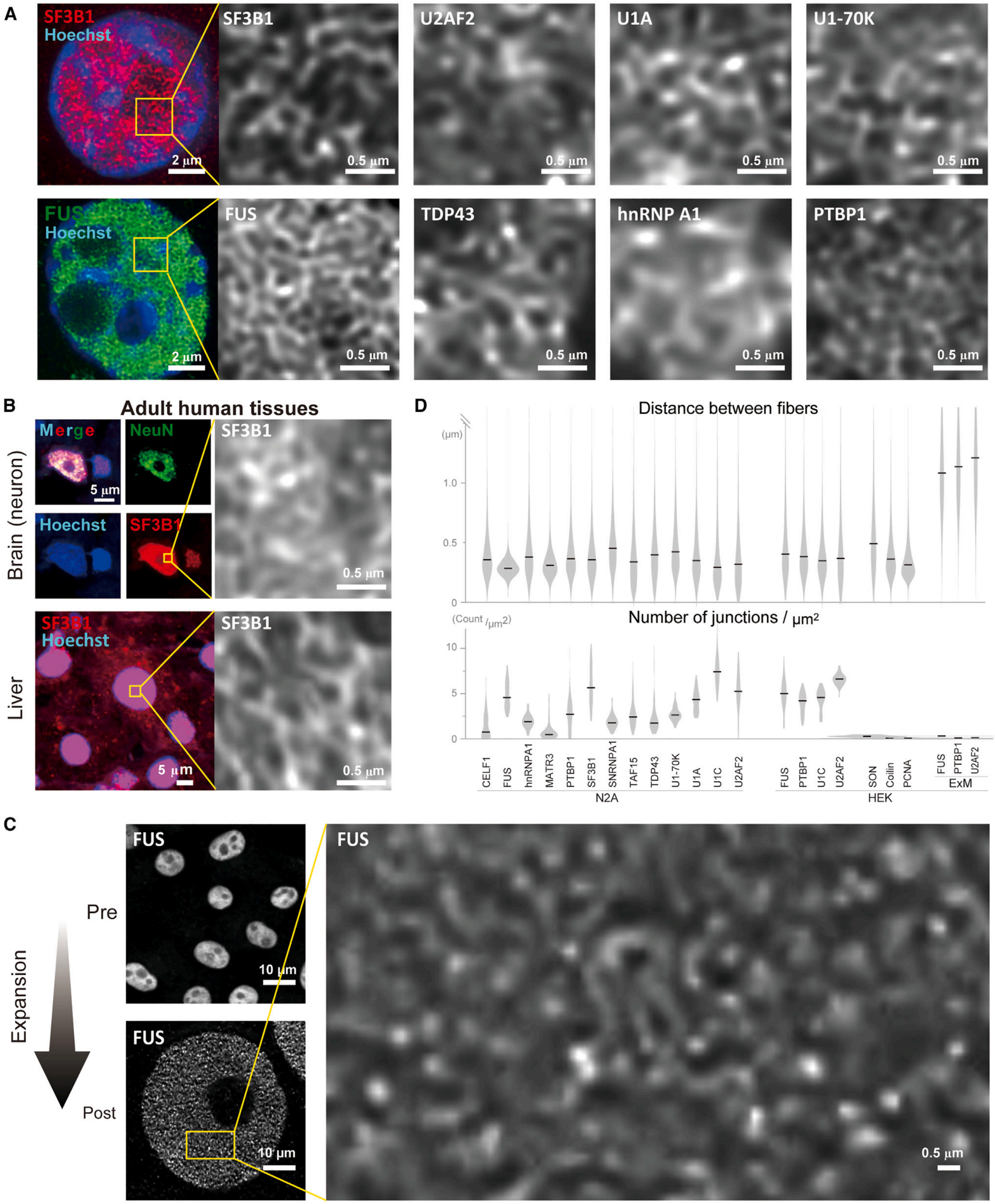
INTRODUCTION

The eukaryotic nucleus is a highly organized structure in which DNA, RNA, and proteins are distributed in a spatiotemporal manner.[1] Heavily compacted DNA is locally loosened by releasing chromatin, and a large transcription complex, including RNA polymerase (RNAP), initiates transcription. Nascent RNA is processed and typically transported through the nucleoplasm to the cytoplasm with the help of specific machinery. The nucleus contains several membrane-less compartments, such as nucleoli and nuclear speckles, where specific proteins and nucleic acids are concentrated to perform their functions.[2] These machinery and compartments are assembled through a dynamic network of multivalent interactions, including protein-protein, protein-nucleic acid, and nucleic acid-nucleic acid interactions.

Splicing is an intricate process catalyzed by a multi-megadalton machinery, the spliceosome. It comprises five small nuclear ribonucleoproteins (U-snRNPs) with hundreds of proteins[3] that assemble on pre-mRNAs and recognize splicing *cis*-elements. Accessory RNA-binding proteins (RBPs) modulate spliceosome activity by binding to specific sites in pre-mRNA in cooperation or competition with spliceosomal components.[4] Several lines of evidence suggest that nuclear architecture is involved in splicing regulation. Distinct nuclear compartments enriched in specific accessory RBPs were enriched in specific sets of alternatively spliced mRNA isoforms.[5,6] The peripheral and central regions of the nucleus are enriched in differently spliced isoforms.[7] Specific nuclear structures are presumably associated with splicing regulation; however, these entities remain unclear.

Recent studies have shown the role of liquid-liquid phase separation (LLPS) in RNA metabolism.[2] RBPs are highly enriched in intrinsically disordered regions (IDRs) that drive LLPS. RNAs alone also exhibit an LLPS activity in a sequence-dependent manner.[8] IDRs are characterized by low sequence complexity and a lack of defined 3D structures, such as domains.[9] The IDRs of RBPs promote the formation of membrane-less compartments such as paraspeckles and nuclear speckles through their LLPS activities.[2,10] Mutations that cause neurodegenerative diseases are often found in IDRs of RBPs and promote the formation of pathogenic aggregates through their dysregulated LLPS.[11] These mutations also induce aberrant splicing associated with neurodegeneration even in cells without aggregates,[12,13] suggesting a functional relationship between splicing and the LLPS of RBPs. However, the underlying mechanisms and compartments responsible for splicing processes remain under investigation.

In this study, we analyzed the subcellular distribution of various RBPs using microscopy strategies with resolutions



(legend on next page)

beyond the optical diffraction limit, including spinning-disk confocal microscopy, stochastic optical reconstruction microscopy (STORM), and expansion microscopy (ExM).[14,15] We observed that multiple RBPs were distributed in meshworks throughout the nuclei of human and mouse cells. *In silico, in vitro*, and *in cellulo* analyses have revealed that core and accessory RBPs in splicing have charged and uncharged IDRs, respectively, which separate from each other to develop distinct RBP meshworks. Crosslinking and immunoprecipitation (CLIP)-seq analysis showed that these meshworks mutually exclusively associate with RNA to regulate alternative splicing. Furthermore, optogenetic enhancement of the meshwork of MATR3, an RBP associated with amyotrophic lateral sclerosis (ALS),[16] caused aberrant splicing, particularly in genes involved in neurodegeneration.

RESULTS

Identification of fibrous networks formed by RBPs in the nuclei of human and mouse cells

During the microscopic analysis of the cellular distribution of RBPs, we noticed that various RBPs formed fibrous networks in the nuclei (Figures 1A and S1A). We observed that both core components of the spliceosome (SF3B1, SNRPA1, U1A, U1C, U1-70K, and U2AF2) and accessory RBPs (CELF1, FUS, hnRNP A1, MATR3, PTBP1, TAF15, and TDP43) were distributed as networks of short, fibrous structures with small patches in the nuclei of N2A and HEK293 cells. At low magnification, the structures appeared to consist of small spots (Figure 1A, color). However, high-magnification analysis using spinning-disk microscopy (Figure 1) and STORM (Figure S1B, FUS) showed reticular structures. A similar MATR3 structure was recently reported in AML12 mouse hepatocytes using stimulated emission depletion (STED) microscopy.[17] An RBP, SON, a marker protein for nuclear speckles, formed large condensates at low magnification and fibrous structures by spinning-disk confocal microscopy in HEK293 cells (Figure S1A), as was shown in a recent article.[18] The fibrous structure of U1C was conspicuous in the nuclear speckles, consistent with the concentrated distribution of U1 snRNP in the nuclear speckles. In contrast to the RBPs, coilin, the component of Cajal bodies, and proliferating cell nuclear antigen (PCNA), the nuclear protein associated with DNA replication, are diffusely distributed throughout the nucleoplasm, forming small speckles but not fibrous structures (Figure S1A), as previously described.[19–21] Live imaging showed that EGFP-tagged MATR3 and EGFP-tagged PTBP1 also formed fibrous networks in the nuclei of N2A cells (Figure S1C). We confirmed that the expression of EGFP alone did not show such fibrous structures, even when stained with an anti-GFP antibody (Ab) (Figure S1D). Analysis of the brain, heart, skeletal muscle, and liver in human and mouse also showed that SF3B1 formed fibrous structures (Figures 1B, S1E, and S1F). These meshworks spread throughout the nuclei, avoiding dense chromatin regions such as nucleoli. These results suggested the ubiquitous formation of meshworks by RBPs in the nuclei of mammalian cells.

We further performed ExM, in which cells are physically expanded in an isotropic fashion to achieve high-resolution analysis.[14,15] ExM revealed enlarged fibrous structures of FUS, PTBP1, and U2AF2 in the nuclei of HEK293 cells (Figures 1C and S1G), as expected. ExM resulted in a ∼5-fold increase in the nuclear size (Figure S1H). Consistently, the median distances between fibers and the median branch lengths were increased several-fold, and the median distribution densities of branches and junctions were decreased (Figures 1D and S1J). ExM also confirmed that EGFP-tagged MATR3, but not EGFP alone, formed fibrous structures as observed in Figures S1B–S1D (Figure S1G, bottom images and right graphs). Additionally, we observed that the distances (299–501 nm), the branch lengths (41–142 nm), and the distribution densities (branches, 2.3–13.9 counts/μm^2; junctions, 0.4–6.2 counts/μm^2) varied among RBPs (Figures 1D and S1J).

Transcription-dependent formation of the RBP meshworks

The next question is how RNA is associated with the RBP meshwork. We introduced 5-bromouridine 5′-triphosphate (BrUTP) into HEK293 cells and visualized the extending RNA in the nucleus (Figure S2A). At 10 min after the introduction of BrUTP, nascent RNA appeared as spots pointing to the RNAPII transcription sites. The RNA spots then extended, forming fibrous structures similar to those of the RBP meshwork. Co-immunostaining analysis showed that the nascent RNA meshwork highly overlapped with the meshworks of core spliceosomal components (SF3B1, U2AF2, and U1C) but less with those of accessory RBPs (FUS, TAF15, PTBP1, CELF1, MATR3, and hnRNPA1) at 180 min (Figures 2A and S2B). Similarly, the meshworks of the core spliceosomal components preferentially overlapped with other core spliceosomal components but less with accessory RBPs (Figures 2A and S2B). These results suggest that core spliceosomal components and accessory RBPs develop distinct meshworks in the nucleus. Although MATR3 and SF3B1 form the most distinct pair of meshworks, they were adjacent and intertwined, suggesting their close relationship in RNA processing (Figure 2A). Additionally, staining of transcribing RNAPII with

Figure 1. Various RBPs form fibrous networks in the nuclei of human and mouse cells

(A) Formation of meshworks by various RBPs in the nuclei of N2A cells. Cells were stained with the antibodies (Abs) specific to the indicated RBPs and Hoechst.

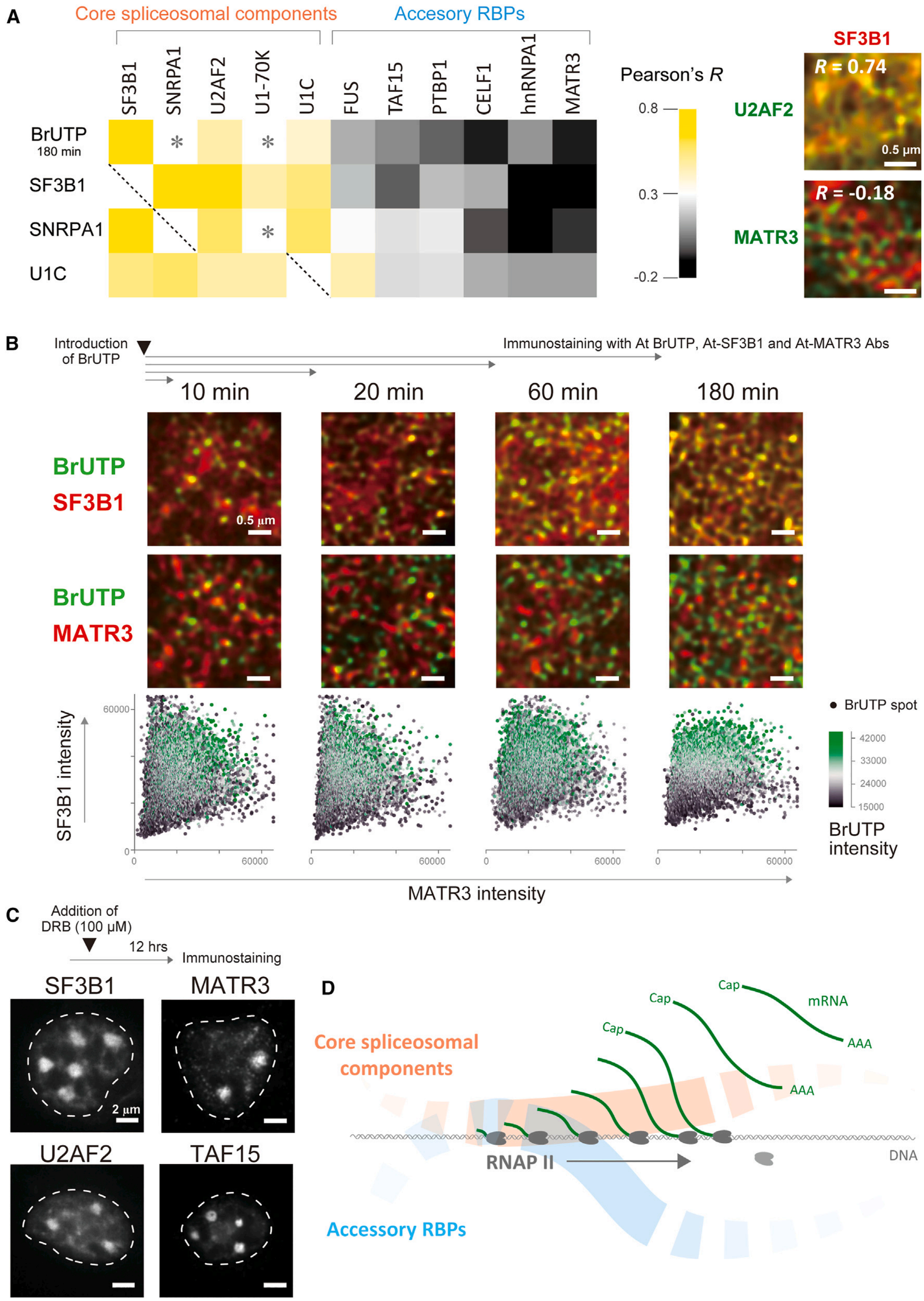
(B) The SF3B1 meshworks in the nucleus of a neuron in the human brain and that of a hepatocyte in the human liver. The sections were stained with anti-SF3B1 Ab and Hoechst.

All images in (A and B) were acquired using the spinning-disk microscopy, SpinSR10.

(C) The FUS meshwork imaged by expansion microscopy. Pre-expanded cells (left upper) and post-expanded cells (left bottom and right) stained for FUS were imaged by confocal microscopy.

(D) The distances between fibers (upper graph) and the distribution densities of junctions (number of junctions/μm^2 nuclear area; lower graph) of the indicated RBP meshworks in N2A and HEK293 cells. Medians are indicated by black bars. "ExM" denotes the post-expanded HEK293 cells.

See also Figure S1.



(legend on next page)

anti-phosphorylated Ser2 RNAPII Ab and of mRNA with anti-5′ cap Ab showed that both distributions overlapped with the SF3B1 meshwork and less frequently with the MATR3 meshwork, although their overlaps were less than that of BrUTP (Figures S2C and S2D). Similarly, the 5′ cap staining overlapped with the U2AF2 meshwork but not with the TAF15 meshwork (Figure S2D). These results suggest that the meshwork of core spliceosomal components is broadly associated with nascent RNA in transcription.

Time course analysis of the BrUTP labeling showed that RNA spots were present on the meshworks of SF3B1 and MATR3 at 10 min (Figure 2B). Thereafter, nascent RNAs progressively overlapped with the SF3B1 meshwork and less with the MATR3 meshwork, along with their extension (Figure 2B). Similarly, RNA spots were present on the meshworks of U2AF2 and PTBP1 at 10 min (Figure S2E), while the nascent RNA mostly overlapped with U2AF2 at 180 min (Figure 2A). Thus, the binding of core spliceosomal components gradually became predominant within minutes to hours. Although splicing is a highly variable kinetic process, a previous study showed that it also takes minutes to hours to complete,[22] suggesting that the progression of the association of core spliceosomal components occurs during the splicing process.

Next, the cells were treated with 5,6-dichlorobenzimidazole riboside (DRB), an RNAPII inhibitor, to eliminate nascent RNA. DRB treatment dissipated the SF3B1, U2AF2, TAF15, and MATR3 meshworks, transforming them into large droplets (Figure 2C). Live imaging analysis of cells expressing EGFP-MATR3 also showed that DRB treatment gradually dissipated the meshwork and formed large droplets in 120 min (Figure S2F). Upon the removal of DRB, the large droplets gradually reverted to small patches to form meshworks. These results demonstrate the essential requirement of active transcription for the formation of the RBP meshwork. In addition, the DRB-mediated transformation into spherical large droplets implied the possible contribution of a phase separation mechanism in creating the meshworks because the analyzed RBPs possess phase separation capacities through their IDRs.[23–25] Indeed, the bleached fluorescence was rapidly recovered in both the meshworks and the large droplets composed of EGFP-MATR3 (Figure S2G).

Classification of IDRs in RBPs

Our analysis showed the unique ability of RBPs to transit between fibrous structures and droplets. Emerging evidence suggests that most RBPs contain IDRs that drive the self-enrichment in cells.[2] To explore the specific features of IDRs that induce the formation of the fibrous structures and droplets of RBPs, we classified IDRs in all the human proteins in an unsupervised manner (Figure 3A).

We first extracted the non-domain regions containing IDRs in all human proteins (Figure 3A), as the accurate prediction of IDRs is still under investigation.[31] Domain annotations were obtained from the Conserved Domains Database (CDD), Pfam, Smart, and Clusters of Orthologous Gene (COG) databases.[32] The extracted non-domain regions were divided into 80 amino acid (aa) bins. The principal component analysis of the 40,133 bins using their aa composition, hydrophobicity, charge, and disorder tendencies (IUPred score)[33] (Table S1) identified three components that accounted for 18.2%, 12.5%, and 7.7% of the total variance. Component 1 contained a disorder tendency (mean IUPred score) and the mean and variance of hydrophobicity, suggesting the role of aa hydrophobicity in forming IDRs. Components 2 and 3 contained the variance and mean charges, respectively, implying the importance of charge properties in differentiating between the IDRs (Figure S3A). The K-means clustering algorithm classified the 40,133 bins into six clusters (Figure S3B). Gene Ontology (GO) analysis showed that four clusters with high IUPred scores were enriched in transcription factors (Figure S3B; Table S2). In addition, the two clusters with the lowest and highest charge variability were significantly enriched in mRNA processing and splicing (blue and red clusters in Figure S3B, respectively).

We next restricted the non-domain region analysis to RBPs. The 2,040 bins in the 521 genes with GO terms of RNA binding and processing were classified into five clusters by K-means clustering (Figures 3B and S3C; Tables S1 and S3). The five clusters are herein called clusters 1–5. The aa compositions and repetitive elements were key determinants of the five clusters (Figures 3C and S3D). The regions in clusters 1–4 had high IUPred scores (Figure 3B), indicating that these regions were IDRs. Furthermore, each cluster was enriched in specific RNA-binding domains and specific speckles (Figure 3D; Table S3). RBPs in cluster 1 were enriched in stress granules and paraspeckles, whereas RBPs in cluster 3 were enriched in nuclear speckles. By contrast, RBPs in cluster 5 were excluded from the speckles, as expected from their low IUPred scores.

As in the IDRs of all human proteins (Figure S3B), the IDRs of RBPs were also differentiated into two major categories by charge variability: constant charge (clusters 1 and 2) and highly variable charges (clusters 3 and 4) (Figure 3B). In addition to the paucity of charge variability within an IDR, RBPs in clusters

Figure 2. Transcription-dependent formation of the RBP meshworks

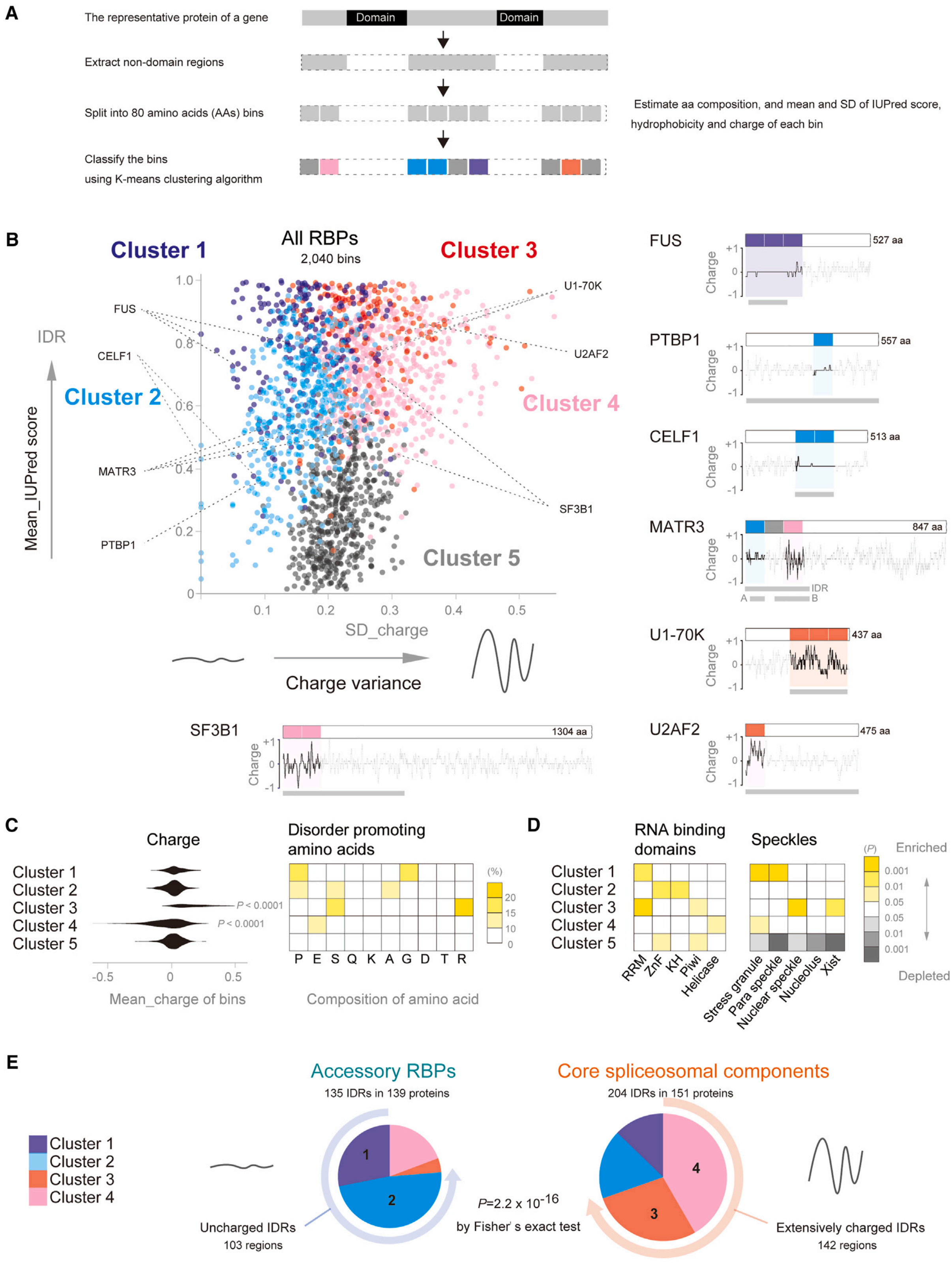
(A) Heatmap showing the overlaps of the meshworks of nascent RNA (BrUTP), SF3B1, SNRPA1, and U1C with eleven RBP meshworks. The indicated meshworks were co-immunostained, and their co-localization was estimated by calculating Pearson's correlation coefficient (*R*) of the intensities composing the two meshworks using the Coloc2 function of Fiji software. * denotes staining with a high background, which prevented the estimation of an *R* value. Representative images of the co-immunostained meshworks are shown in the right panels.

(B) Overlaps between the nascent RNA meshwork (green), the SF3B1 meshwork (red), and MATR3 meshwork (red) in HEK293 cells. Immunostaining was performed at the indicated time after the introduction of BrUTP. (Bottom graphs) Each dot represents a single BrUTP spot, and its intensity is indicated by a heatmap. The signal intensities of SF3B1 and MATR3 on each BrUTP spot are plotted on the y and x axes, respectively.

(C) Transformation of the RBP meshworks into droplets by transcription arrest. N2A cells were treated with DRB (100 μM) for 12 h to arrest transcription and were immunostained with the indicated Abs. A white, broken contour outlines the nucleus.

(D) Schematic showing the localization of the RBP meshworks, nascent RNA, and RNAPII during the progression of transcription.

See also Figure S2.



(legend on next page)

1 and 2 in the first category carried mostly uncharged IDRs (Figure 3C, left). By contrast, RBPs in clusters 3 and 4 in the second category carried positively and negatively charged IDRs on average, respectively (Figure 3C, left). GO analysis showed that the first category was enriched in RBPs involved in the regulation of mRNA processing, whereas the second category was enriched in RBPs in the core spliceosomal complex (Figure S3E). Consistently, out of 135 IDRs in the 139 accessory RBPs listed in a previous article,[34] 103 IDRs (76.3%) were classified into the first category with uncharged IDRs, whereas 142 IDRs (69.6%) out of 204 IDRs in the 151 core spliceosomal components annotated by the Human Genome Organization (HUGO) Gene Nomenclature Committee[35] were classified into the second category with charged IDRs (Figure 3E; Table S4). In summary, core spliceosomal components and accessory RBPs harbor distinct classes of charged and uncharged IDRs, respectively.

Blending and separating RBP condensates by charged/uncharged IDRs

Given the remarkable differences in the physicochemical properties of IDRs between core spliceosomal components and accessory RBPs, we investigated how IDRs in the different clusters formed distinct meshworks. We selected FUS for cluster 1, PTBP1 and CELF1 for cluster 2, U1-70K and U2AF2 for cluster 3, and SF3B1 for cluster 4 as representative RBPs (Figure 3B, right). We also selected MATR3, which had IDRs in clusters 2 and 4. We generated purified recombinant proteins of these RBPs (Figure 3B), which were fused with AcGFP or mCherry. We also generated two AcGFP-fused proteins comprised of the first and second MATR3 IDRs in clusters 2 and 4, respectively (MATR3-A and MATR3-B) (Figure 3B). For PTBP1 and U2AF2, we also made AcGFP- or mCherry-fused mutant proteins lacking the IDRs (labeled as ΔIDR).

Confocal imaging showed that all IDR-containing proteins formed droplets or fibrous aggregates, which are herein collectively called condensates, under *in vitro* phase separation conditions (Figures 4A and S4A–S4D). The condensates of proteins with uncharged IDRs (FUS, PTBP1, CELF1, and MATR3-A) resisted the increased ionic strength. By contrast, high salt concentrations dissipated the condensates of charged IDRs and reduced their numbers (U1-70K, U2AF2, SF3B1, and MATR3-B) (Figures 4A, S4A, and S4D). In addition, PTBP1-ΔIDR, U2AF2-ΔIDR, and CELF1-ΔIDR markedly reduced the numbers of condensates (Figures 4B, S4B, and S4C) compared with those of their full-length proteins, indicating the requirement of IDRs for their condensation. Thus, the condensates formed by the uncharged and charged IDRs were resistant and vulnerable to high salt concentrations, respectively.

Next, we examined whether proteins in different clusters form blended or separated condensates *in vitro*. Two proteins labeled with either AcGFP or mCherry were mixed under phase separation conditions, and each condensate's green and red fluorescence intensities were analyzed by confocal imaging. For example, U2AF2-AcGFP and SF3B1-mCherry formed well-blended droplets (Figure 4C, left). The correlation coefficient (R) between the green and red fluorescence intensities was 0.79. By contrast, PTBP1-AcGFP and SF3B1-mCherry formed separated droplets with an R value of -0.63. We examined 33 different combinations of purified proteins and calculated the R values (Figures 4C, right and S4E). We found that proteins with uncharged IDRs (clusters 1 and 2) blended well with each other. Similarly, proteins with charged IDRs (clusters 3 and 4) blended well with each other. However, proteins with uncharged IDRs did not blend with those with charged IDRs, suggesting that charged and uncharged IDRs form separated condensates. The MATR3 IDRs, which contain both uncharged and charged regions (clusters 2 and 4, respectively), blended well with PTBP1 and CELF1 (cluster 2) and weakly with U1-70K (cluster 3) but not with the other RBPs.

We further investigated the role of IDRs in forming the RBP meshwork within the nucleus. We constructed expression vectors of EGFP-tagged MATR3 (wild type [WT]) and its three mutants, where the IDR was deleted (ΔIDR), replaced with that of PTBP1 (PTBP1-IDR), and replaced with that of SF3B1 (SF3B1-IDR) (Figure 4D, left). These vectors were introduced into HEK293 cells along with the MATR3-mCherry expression vector, and the distribution of the expressed proteins was analyzed. We observed that all these proteins were localized to the nucleus (Figure S4I). ΔIDR showed the lowest overlap with MATR3-mCherry (Figure 4D) and little overlap with endogenous SF3B1 and U2AF2 (Figure S4F), suggesting the requirement of the IDR for participating in the meshworks. The meshwork of PTBP1-IDR overlapped well with that of MATR3-mCherry, whereas that of SF3B1-IDR did not (Figure 4D), as observed in the recombinant protein experiments (Figure 4C). SF3B1-IDR, however, overlapped well with MATR3 lacking the uncharged IDR region (ΔA), suggesting the importance of charged and uncharged IDRs for the distribution of MATR3. We also examined the distributions of AcGFP/mCherry-tagged U2AF2 (U2_WT) and its three IDR mutants, where the IDR was deleted (U2_ΔIDR), replaced with that of MATR3-A (U2_MA3A-IDR), and replaced with that of SF3B1 (U2_SF3B1-IDR) (Figure S4G, left). We observed that

Figure 3. Classification of IDRs in RBPs

(A) Schematic showing the process of the unsupervised classification of IDRs.

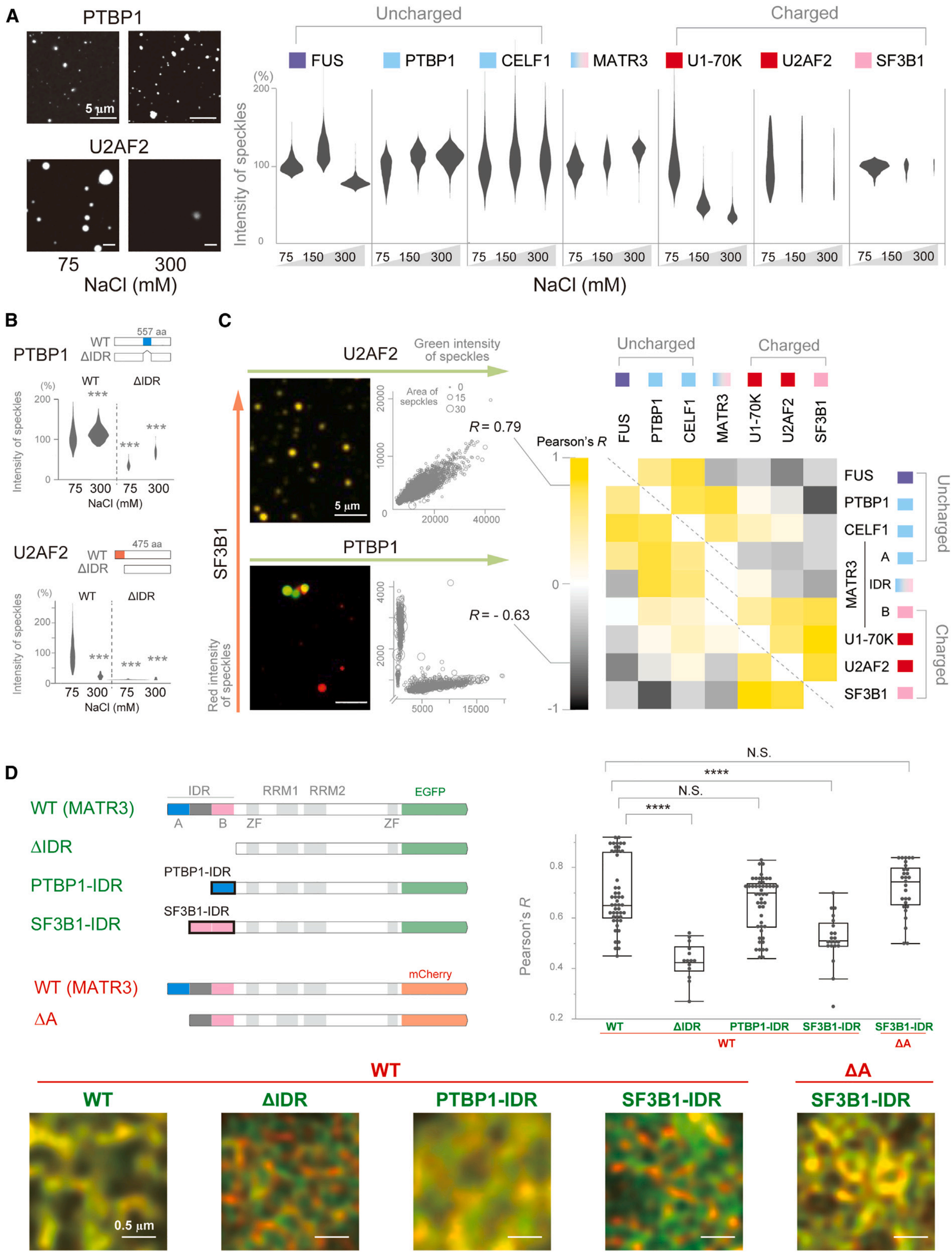
(B) Five clusters were identified by K-means clustering of non-domain regions in human RBPs. Charge variance (standard deviation [SD] of charges) and disorder tendency (average IUPred score) of an 80-aa bin are plotted on the x and y axes, respectively. Clusters 1–5 are color-coded. Each dot represents an 80-aa bin in a non-domain region of an RBP. Charge distributions of representative RBPs are indicated in right panels. The blocks in the proteins indicate the 80-aa bins in non-domain regions and are color-coded as in the graph. The gray bars below the charge plots denote the regions of the recombinant proteins used in Figures 4A–4C.

(C) Mean charges (left) and the ratios of disorder-promoting amino acids of the 80-aa bins in the five clusters. p values were calculated with the Steel-Dwass test.

(D) Enrichments of the RNA-binding domains (left) and of RBPs in the speckles (right) in the five clusters. RBPs concentrated in each speckle were obtained from the proteome analyses in previous reports.[26–30] p values were calculated using Fisher's exact test.

(E) The accessory RBPs were preferentially classified into clusters 1 and 2 (76.3%), whereas the core RBPs were classified into clusters 3 and 4 (69.6%). p values were calculated by Fisher's exact test.

See also Figure S3 and Tables S1, S2, S3, and S4.



(legend on next page)

the U2_WT and U2_SF3B1-IDR were mostly localized to the nucleus, while U2_ΔIDR and U2_MA3A-IDR were distributed in both the cytoplasm and nucleus (Figures S4G and S4H). Similar to the ΔIDR of MATR3, U2_ΔIDR showed the lowest overlap with the meshwork of U2_WT (R = 0.525) in the nucleus, although the separation was less noticeable compared with the separation of the MATR3 proteins (R = 0.425, Figure 4D). U2_MA3A-IDR (uncharged IDR) was still separated from U2_WT (charged IDR), whereas U2_SF3B1-IDR (charged IDR) retained the overlap with U2_WT (charged IDR, Figure S4G). Overall, our analysis revealed the requirement of IDRs for the distribution of RBP meshworks, in which the separation of charged and uncharged IDRs plays a role.

The role of IDRs in RBPs in RNA binding

Next, we explored the role of RBP meshwork formation in RNA binding using CLIP-seq. We prepared expression vectors for FLAG-tagged WT MATR3 and its deletion mutants lacking RRM1 (ΔRRM1), RRM2 (ΔRRM2), and IDR (ΔIDR) (Figure 5A). These vectors were introduced into HEK293 cells (Figures S5A and S5B), and CLIP-seq was performed using the anti-FLAG Ab. CLIP-seq of endogenous MATR3 (Endo) was used as a control. We confirmed that the reads of Endo-CLIP and WT-CLIP were abundantly distributed in intronic regions (Figure 5D) and enriched in CU-rich motifs (Table S5), as previously reported.[36,37] Comparison of read coverage for contiguous 10 kb bins of the whole genome revealed that the reads of WT-, ΔRRM1-, and ΔRRM2-CLIPs were distributed similarly to those of Endo-CLIP across the transcriptome (R = 0.75, 0.83, and 0.81, respectively). By contrast, the read distribution of ΔIDR-CLIP was noticeably deranged (R = 0.56) despite the retention of the two RRMs (Figures 5B, 5C, S5C, and S5D). The metagene analysis showed that the reads of ΔIDR were markedly accumulated in exons and promoter regions compared with WT (Figure S5E).

We similarly performed CLIP-seq of the WT and ΔIDR U2AF2 (U2_WT and U2_ΔIDR, respectively). Since U2_ΔIDR was partially localized to the cytoplasm (Figures S4G and S4H), we performed CLIP-seq of nuclear lysates to specifically analyze the effects of the separated meshworks on RNA binding. We observed that the reads of U2_WT were accumulated toward 3′ splice sites in introns (Figure S5F), representing the close association of U2AF2 with polypyrimidine tracts.[24] The reads of U2_ΔIDR were, however, decreased in those regions. Additionally, the overall read distribution of U2_ΔIDR was mildly affected (Figure S5F), consistent with the moderate separation of U2_ΔIDR from the meshwork of U2_WT (Figure S4G). Taken together, both MATR3 and U2AF2 mutants lacking their IDRs showed disturbed localization to the meshworks of the WT proteins (Figures 4D and S4G) and did not properly bind to target segments in nascent RNA. These results suggest that RBP meshworks are the regions where RBPs interact with pre-mRNAs.

Next, we performed CLIP-seq of endogenous MATR3, PTBP1, and SF3B1 using naive and *Matr3*-knockdown N2A cells to understand how the absence of one meshwork affects the RNA-binding of the other interacting meshwork. MATR3 was exclusively localized to the nucleus, forming a meshwork (Figure S1A). We confirmed that the reads of MATR3-CLIP and PTBP1-CLIP were preferentially distributed in intronic regions (Figure S5G) and enriched in CU-rich motifs (Table S5), as previously reported.[36,37] SF3B1 recognizes a branch site with a degenerative UNA motif to promote the splicing of a 3′ splice site by the spliceosome.[38] Consistently, the reads of SF3B1-CLIP were accumulated just upstream of 3′ splice sites, where branch sites are located, and were enriched in UNA motifs and a splice site-like motif, AGGU (Figures 5F and S5I). We classified MATR3-binding peaks (>280 nt) identified with the peak detection algorithm, MACS,[39] into three categories based on their length (Figure 5E). PTBP1-CLIP reads were also accumulated at the MATR3 peaks, as previously reported.[36] *Matr3*-knockdown did not affect the distribution of PTBP1-CLIP reads at the MATR3 peaks. By contrast, SF3B1-CLIP reads were reduced around the MATR3 peaks, which, however, were increased by *Matr3*-knockdown. This reduction in SF3B1-binding was more evident around larger MATR3 clusters (Figure 5E) and also in deep introns, where MATR3 preferentially binds to suppress cryptic splice sites (Figures 5D and 5F).[37] These results suggest that MATR3-condensation suppresses the association of SF3B1 with RNA in a dose-dependent manner.

We similarly analyzed our previously reported CLIP data for FUS and the components of U1 snRNP, U1C, and U1-70K (DDBJ: DRR171207, DRR171208, DRR171206, and DRR494391).[40]

Figure 4. Blend and separation of RBP condensates by charged/uncharged IDRs

(A) *In vitro* droplet formation of the recombinant proteins of the indicated RBPs at different salt concentrations. Representative images are shown in the left panels. The width of the violin plot represents the number of droplets. AcGFP- or mCherry-fused recombinant proteins (10 nM) were added to the droplet formation buffer. The regions of the RBPs used to construct the recombinant proteins are indicated in Figure 3B (gray bars in the right panels). The RBPs are color-coded by their IDR clusters, as in Figure 3B.

(B) Droplet formation of the recombinant wild-type (WT) proteins and their mutants lacking the IDR (ΔIDR) for PTBP1 (top) and U2AF2 (bottom) at different salt concentrations.

(C) Blend or separation of two RBPs in *in vitro* droplet formation assays. The AcGFP-fused RBP was mixed with the mCherry-fused RBP and added to the droplet formation buffer (75 mM NaCl). Representative images and scatterplots of the green and red fluorescence intensities of each droplet are shown in the left panels. Pearson's correlation coefficients (R) between the green and red fluorescent intensities are shown in the right heatmap. The RBPs are color-coded by their IDR clusters, as in Figure 3B.

(D) Blend or separation of two meshworks in cells. HEK293 cells expressing indicated combinations of EGFP- or mCherry-tagged WT MATR3 or mutant MATR3 (shown in the left upper corner) were observed by the spinning-disk confocal microscopy. Representative images are shown below. Pearson's correlation coefficients (R) of the green and red signal intensities are shown in the right graph. The box plot shows the interquartile range (boxes), the median (central band), and the minimum and maximum except for the outliers at the ends of whiskers. Each dot represents a single z stacked visual field. ****p < 0.0001, N.S., no significance by the Steel test compared with WT.

See also Figure S4.

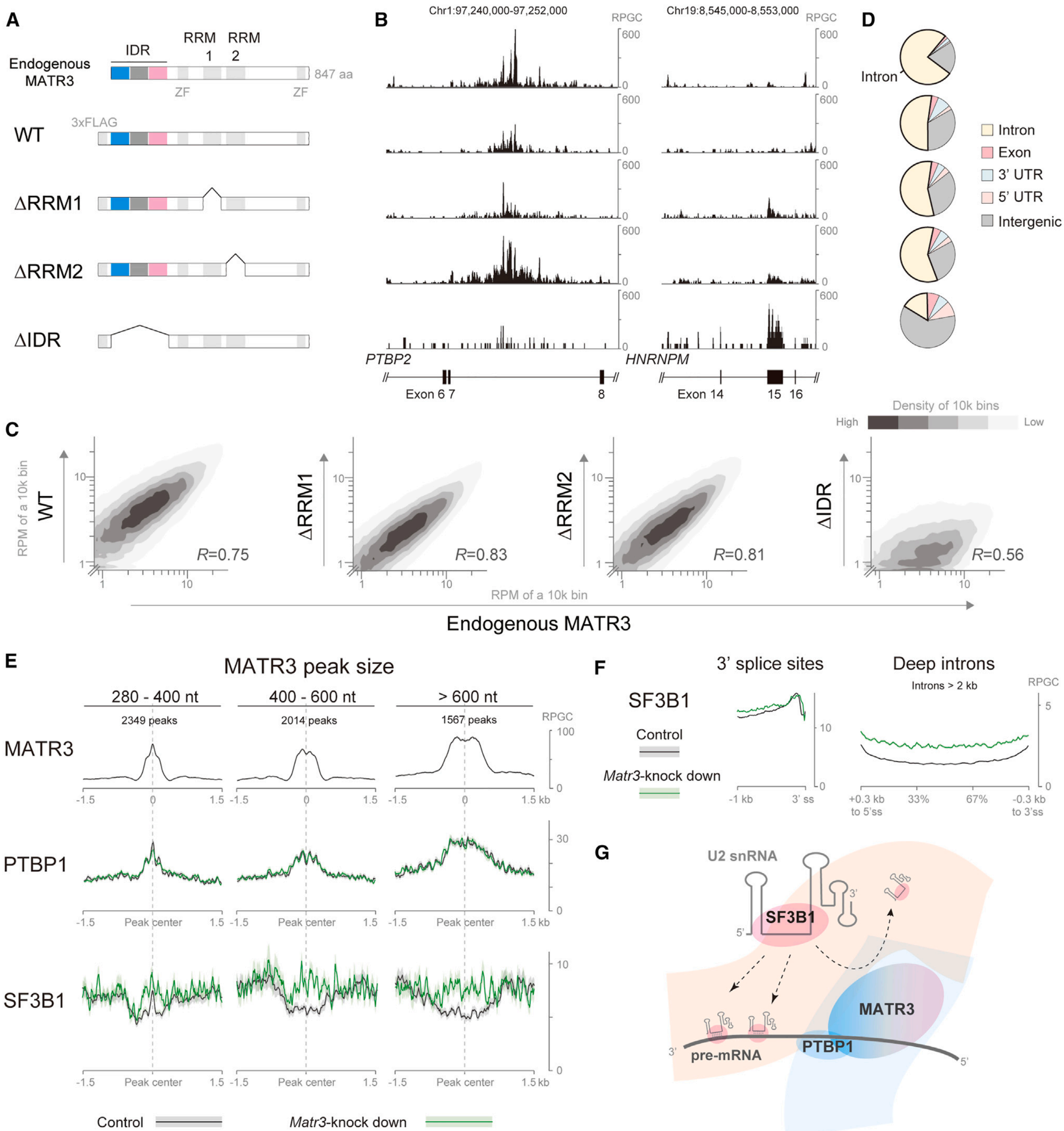


Figure 5. The role of IDRs in RBPs in RNA binding

(A) Protein structures of WT MATR3 and its deletion mutants (ΔRRM1, ΔRRM2, and ΔIDR) expressed in HEK293 cells for CLIP-seq analysis. CLIP-seq of endogenous MATR3 was performed using naive HEK293 cells and anti-MATR3 Ab. The other MATR3-CLIP-seqs (WT, ΔRRM1, ΔRRM2, and ΔIDR) were performed using HEK293 cells expressing the indicated proteins and anti-FLAG Ab.

(B) Distributions of CLIP-seq reads of MATR3 indicated in (A) on chr1:97,240,000–97,252,000 (GRCh37/hg19) within the *PTBP2* gene and chr19:8,545,000–8,553,000 within the *HNRNPM* gene.

(C) Correlation of read densities of 10-kb bins across the whole human genome between the endogenous MATR3-CLIP and the other MATR3-CLIPs (WT, ΔRRM1, ΔRRM2, and ΔIDR). The densities of 10,000-nt bins are shown in heatmaps.

(D) Pie charts showing the peak distributions of MATR3-CLIP-seqs indicated in (A) in HEK293 cells.

(legend continued on next page)

FUS forms a complex with U1 snRNP by directly interacting with U1 snRNA (Figure S5L).[41,42] FUS and U1C have uncharged IDRs, whereas U1-70K has a charged IDR (Figure S5K). We observed that the CLIP reads of U1C and U1-70K accumulated around the FUS peaks, as expected. *Fus*-knockdown did not affect the distribution of U1C-CLIP reads, but increased the distribution of U1-70K-CLIP reads (Figure S5K). This is in accordance with the notion that the uncharged IDR in FUS does not compete for the uncharged IDRs of U1C but competes for the charged IDRs in U1-70K.

Both *Matr3*-knockdown and *Fus*-knockdown experiments showed that the local accumulation of the RBP with an uncharged IDR (MATR3 and FUS) prevents the binding of the RBP with a charged IDR (SF3B1 and U1-70K, respectively) but not of the RBP with an uncharged IDR (PTBP1 and U1C, respectively) to pre-mRNA (Figures 5G and S5L). To summarize, uncharged and charged IDRs mutually exclude each other from binding to pre-mRNA.

Condensation and dissipation of MATR3 meshwork affect alternative splicing of distinct gene sets

Finally, we investigated the role of meshwork formation in alternative splicing regulation. We used a two-module optogenetic system Corelets to induce protein condensation around a ferritin core complex via a blue light-inducible improved light-induced dimer (iLID) switch (Figure 6A).[43] HEK293 cells expressing MATR3 fused to mCherry and SspB (MATR3-mCherry-SspB) along with EGFP fused to iLID and FTH1 (iLID-EGFP-FTH1) were repeatedly exposed to blue light for 12 h to analyze the changes in gene expression and alternative splicing. The light-induced condensation of MATR3-mCherry-SspB reinforced the formation of a meshwork in HEK293 cells (Figure 6A, middle). Additionally, we prepared two lines of control cells in which condensation was not induced: non-irradiated cells expressing MATR3-mCherry-SspB and iLID-EGFP-FTH1 and irradiated cells expressing MATR3-mCherry-SspB and EGFP alone. We isolated cells expressing similar levels of MATR3-mCherry-SspB by cell sorting and performed RNA sequencing (RNA-seq) analysis (Figure 6A).

The light-induced condensation of MATR3-mCherry-SspB affected 537 splicing events and 218 gene expressions (Figures 6B and S6B). Previous studies have shown that MATR3 represses alternative splicing by binding to intronic regions around target exons with minimal changes in gene expression.[25,36] Consistently, splicing repression was the most prevalent event induced by the condensation (Figure 6B). WT-MATR3-CLIP analysis showed that MATR3 was recruited to the repressed regions in cassette exons, exonic regions in the splice site shift events, and retained intronic regions (Figures 6C and S6C). Thus, the condensation of the MATR3 meshwork represses splicing by promoting the direct binding of MATR3 to RNA.

We also performed RNA-seq analysis of *MATR3*-knockdown cells. As expected, *MATR3*-knockdown preferentially activates splicing of the regions, where endogenous MATR3 bound abundantly (Figures 6B, 6C, S6B, and S6C). We found that MATR3-condensation and *MATR3*-knockdown affected the splicing of distinct gene sets (Figure 6D). Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analysis revealed that the splicing events affected by MATR3-condensation but not by *MATR3*-knockdown are significantly concentrated in genes associated with neurodegenerative diseases (Figure 6E).

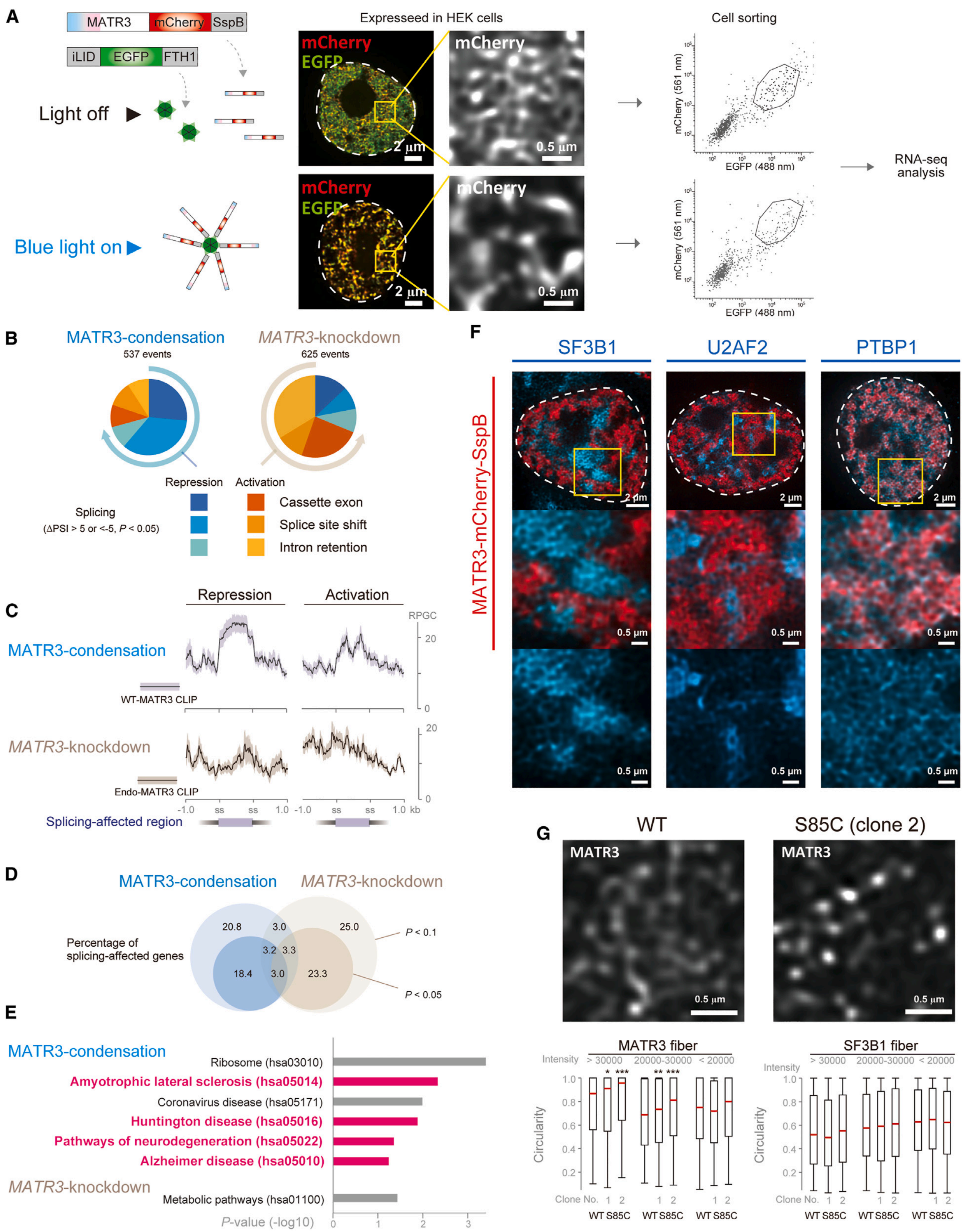
Immunostaining analysis showed that *MATR3*-knockdown disrupted a meshwork, transforming it into small spots (Figure S6D). By contrast, the overexpression of MATR3-mCherry-SspB enhanced the meshwork formation, which was further promoted by the light-induced MATR3-condensation, indicating the importance of the local RBP concentration for the meshwork formation. The condensed MATR3 meshwork extensively excluded endogenous SF3B1 and U2AF2 but not PTBP1 in the nucleus (Figure 6F). The spatial exclusion of SF3B1 and U2AF2 from the MATR3 meshwork, where MATR3 binds to RNA, is likely to contribute to the splicing repression by MATR3-condensation.

In addition, we observed that ALS-causative autosomal dominant mutations of MATR3, S85C, and F115C, which were located in the IDR,[36] also affect the meshwork formation. We introduced the expression vectors of these MATR3 mutants into N2A cells and found that the mutants formed coarse fibrous structures compared with WT MATR3 while retaining the ability to exclude endogenous SF3B1 (Figure S6E). The S85C mutation resulted in focal clusters of short fibers, whereas the F115C mutation resulted in an overall thickened meshwork. The high viscosity of the S85C condensate[44] is likely to lead to the fragmentation of the fibrous structures. Consistent with this, CRISPR-Cas9-mediated hemiallelic introduction of the S85C mutation into N2A cells resulted in the disruption of a fibrous meshwork and the circularization of MATR3 signals (Figures 6G and S6G). We confirmed that the expression of MATR3 was not affected in these cells (Figure S6A). These results suggest that a mutation in an IDR can affect the meshwork formation in addition to the local concentrations of an RBP.

DISCUSSION

In this study, we demonstrate that RBPs recruited to pre-mRNA form meshworks that spread throughout the nucleus to conduct the splicing process. Core spliceosomal components and accessory RBPs regulating RNA processing constitute two

(E) Read distributions of MATR3-CLIP (top), PTBP1-CLIP (middle), and SF3B1-CLIP (bottom) around the MATR3-CLIP peaks. MATR3-CLIP was performed using naive N2A cells. PTBP1-CLIP and SF3B1-CLIP were performed using *Matr3*-silenced (green) or control (black) N2A cells. The standard error is shown as a semi-transparent shade around the average curve.
(F) Read distributions of SF3B1-CLIPs mapped upstream of the 3′ splice sites (ss) of all coding exons and those mapped to the relative positions of deep introns (from 0.3 kb downstream of 5′ ss to 0.3 kb upstream of 3′ ss of all introns longer than 2 kb). The standard error is shown as a semi-transparent shade around the average curve.
(G) Schematic showing the exclusion of SF3B1 but not PTBP1 from a MATR3 cluster on pre-mRNA.
See also Figure S5 and Table S5.



(legend on next page)

distinct meshworks that are adjacently but distinctly distributed, and mutually exclusive on pre-mRNA. The separation of the charged and uncharged IDRs in these RBPs achieves the mutual exclusion. Our analysis revealed that the spatial organization of RBP networks is a functional regulator of RNA splicing.

Tracing of RNAs under transcription by BrUTP showed that RBP meshworks were formed with the extension of nascent RNA. Both core spliceosomal components and accessory RBPs were tethered to nascent RNAs immediately after transcription (Figures 2B and S2E). A previous report identified the role of multivalent RNA-RNA interactions in forming filamentous structures in cytosolic granules.[45] In their report, longer RNA with fewer secondary structures makes mesh-like structures more complex. Because nascent pre-mRNAs include introns and are long, they are likely to induce the formation of the RBP meshwork in the nucleus.

More than 30 years ago, an electron microscopic analysis identified "core filaments," a reticular structure composed of RNA and RBPs in the nucleus.[46] The core filaments comprised $\sim$10 nm fibers that spread throughout the nucleus, which were rapidly degraded by RNase treatment. The core filaments are identical to the RBP meshworks that we observed. However, the existence of core filaments in living cells has long been questioned and is not currently postulated because the filaments appear only after extensive treatment of the samples to remove the chromatin fraction, such as treatments with high salt buffer and DNase.[47] Our analysis identified at least two types of RBP meshworks. One comprises accessory RBPs whose condensation resists increased ionic strength (Figure 4A). A subset of RBP meshworks resistant to the extensive treatment may be detected as core filaments.

IDRs are well known to drive LLPS through multivalent interactions involving the composition and sequence of aa.[48] A recent study showed that transcription factors with a similar alignment of positive and negative charge blocks in IDRs were selectively recruited to the condensate of the transcription factor MED1.[49] This study reveals the importance of the presence or absence of charges on the IDR for splicing regulation. Our classification showed that many transcription factors had uncharged IDRs (Figure S3B). In addition to IDRs with different charge-block patterns, uncharged IDRs in transcription factors may have distinct functions to regulate transcription. Conversely, the subgroups of charged IDRs in RBPs (clusters 3 and 4; Figure 3B) may have different charge patterns that affect RNA processing. Recently, at least half of all transcription factors have been shown to bind to RNA, not only to DNA,[50] suggesting that IDRs in transcription factors and those in RBPs cooperatively or competitively contribute to their selective recruitment to a target RNA.

We found that the MATR3 and U2AF2 mutants lacking the IDRs did not localize to the meshworks of the WT proteins (Figures 4D and S4G) nor properly bind to target segments in nascent RNA (Figures 5C, 5D, and S5C–S5F). In addition, enhancement of the MATR3 meshwork repressed the splicing in the regions where MATR3 binds abundantly to RNA (Figure S6C). These results demonstrate that RBP meshworks are the regions where RBPs interact with pre-mRNAs to process them. We observed that MATR3 clusters spatially excluded the SF3B1 and U2AF2 by separating uncharged and charged IDRs (Figures 4D and 6F). Furthermore, MATR3 clusters prevented SF3B1 from binding to nascent RNAs (Figure 5E), leading to splicing repression. Similar to MATR3, PTBP1 condensates separated from the charged IDRs of core spliceosomal components such as SF3B1 and U2AF2 (Figure 4C), suggesting that multimerized PTBP1 can interrupt the recognition of splicing *cis*-elements by the core spliceosomal components. In addition, FUS, which has an uncharged IDR, prevented the association of U1-70K, which has a charged IDR, with pre-mRNA (Figure S5K). These results suggest that the separation of uncharged and charged IDRs is extensively operational in regulating RNA processing.

It is well known that the mutations in the IDRs of RBPs induce aberrant RNA processing associated with neurodegeneration.[11] Although abnormal LLPS generated by mutated IDRs promotes the formation of toxic aggregates, they are not always found in affected cells,[12,13] suggesting that aggregates are not a unique cause of the aberrant RNA processing. We demonstrated that the artificial enhancement of the MATR3 meshwork induces

Figure 6. Artificial enhancement of the MATR3 meshwork leads to aberrant splicing

(A) (Left) Schematic of the experiment using the Corelet system.[43] MATR3-mCherry-SspB and iLID-EGFP-FTH1 were expressed in HEK293 cells. Blue light-induced iLID-mediated condensation of SspB around the ferritin core (FTH1). (Middle) Light-induced enhancement of the meshwork of MATR3-mCherry-SspB. A white, broken contour outlines the nucleus. (Right) Cells expressing similar levels of MATR3-mCherry-SspB marked by polygons were isolated by cell sorting and subjected to RNA-seq.

(B) Pie charts showing the numbers of affected splicing events by blue light-mediated enhancement of the MATR3 meshwork (MATR3-condensation) and *MATR3*-knockdown.

(C) Read distributions of WT-MATR3 CLIP and Endo-MATR3 CLIP (see Figure 5A) around the differentially spliced sites by MATR3-condensation and MATR3-knockdown, respectively. The standard error is shown as a semi-transparent shade around the average curve.

(D) Venn diagram showing the overlap between the splicing-affected genes by MATR3-condensation and those by *MATR3*-knockdown. The significantly affected splicing events were analyzed ($p < 0.05$ and $p < 0.1$).

(E) KEGG pathway analysis of the genes with the affected splicing events by MATR3-condensation and *MATR3*-knockdown. The pathways associated with neurodegeneration are indicated in red. p values were calculated using Fisher's exact test.

(F) Exclusion of the endogenous SF3B1 and U2AF2 but not PTBP1 from the coarsened MATR3 meshwork in HEK293 expressing MATR3-mCherry-SspB, exposed to blue light. A white, broken contour outlines the nucleus. Yellow rectangles indicate the regions magnified in the lower panels.

(G) Circularity of the meshes in naive N2A cells (WT) and the N2A cells carrying the ALS-associated mutation, S85C, in one allele of *MATR3*. (Upper) Representative immunostaining images. (Lower) Box-and-whisker plots showing the circularity of the fibers. The meshes are divided into three categories based on their intensities in the immunostaining images. The plots show the interquartile range (boxes), the median (central red band), and the minimum and maximum except for the outliers at the ends of whiskers. *p <0.05, **p <0.01, ***p < 0.001 by on the Steel-Dwass test.
See also Figure S6.

aberrant splicing (Figure 6B). MATR3-condensation but not MATR3-knockdown affected the splicing of genes associated with neurodegenerative diseases (Figure 6E), implying a specific role of the meshwork in splicing. Furthermore, a mutation in the IDR of MATR3 disturbed the meshwork (Figure 6G). Although we only analyzed MATR3 mutations and further studies are required, compromised meshwork formations by the mutations in the IDRs of RBPs are likely to be associated with aberrant RNA processing in neurodegeneration.

Limitations of the study

We proposed the model of spatially competitive RBP networks that conduct RNA splicing. Although the CLIP-seq showed that the competition results in differences in RNA-binding status, it remains unclear whether these RBPs directly compete for RNA binding at the individual molecular level. The CLIP-seq and RNA-seq were performed using N2A and HEK cell lines. The use of these cells has advantages in manipulating a specific gene expression and in obtaining sufficient amounts of input materials to generate high-resolution transcriptome maps. However, it will miss many cell-type-specific events. Additionally, the knockdown of a splicing factor could affect the expression and splicing of other genes. Although our comprehensive analyses suggest that the competition between meshworks regulates RNA processing, the changes detected by the CLIP-seq may include both direct and indirect effects of the RBP knockdown. The CLIP-seq of MATR3-deletion mutants showed that the read distribution of ΔIDR was deranged (Figures 5B and 5C). ΔIDR retains the two RRMs, suggesting that the impaired localization of ΔIDR to the meshworks (Figure 4D) disrupts the proper binding to RNA. Alternatively, a conformational change caused by the IDR deletion might affect its binding preference to RNA. Although the recombinant protein of MATR3 tends to form aggregates, *in vitro* analyses for precise protein-RNA interactions and protein folding will provide further insight.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.molcel.2024.07.001.

ACKNOWLEDGMENTS

We acknowledge Ms. Eri Yorifuji at the Division for Medical Research Engineering, Nagoya University Graduate School of Medicine, for technical support of the microscopic analysis. This work was supported by JSPS KAKENHI (grant number 22H04925 [PAGS], 21H02476, 22K19269, JP23K18273, and 23H02794); the Ministry of Health, Labor, and Welfare of Japan (23FC1014); the Japan Agency for Medical Research and Development (JP23ek0109497); and the Intramural Research Grant for Neurological and Psychiatric Disorders of NCNP (5–6).

AUTHOR CONTRIBUTIONS

A.M. designed the experiments. A.M., T.O., and T.H. conducted the experiments. A.M., T.K., and J.-i.T. performed the computational analyses. A.M. and K.O. wrote the paper.

DECLARATION OF INTERESTS

The authors declare no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used Grammarly and DeepL to improve language and readability. After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Received: August 11, 2023
Revised: January 28, 2024
Accepted: July 2, 2024
Published: July 24, 2024

REFERENCES

1. Hafner, A., and Boettiger, A. (2023). The spatial organization of transcriptional control. Nat. Rev. Genet. *24*, 53–68. https://doi.org/10.1038/s41576-022-00526-0.
2. Hirose, T., Ninomiya, K., Nakagawa, S., and Yamazaki, T. (2023). A guide to membraneless organelles and their various roles in gene regulation. Nat. Rev. Mol. Cell Biol. *24*, 288–304. https://doi.org/10.1038/s41580-022-00558-8.
3. Van Nostrand, E.L., Pratt, G.A., Yee, B.A., Wheeler, E.C., Blue, S.M., Mueller, J., Park, S.S., Garcia, K.E., Gelboin-Burkhart, C., Nguyen, T.B., et al. (2020). Principles of RNA processing from analysis of enhanced

CLIP maps for 150 RNA binding proteins. Genome Biol. *21*, 90. https://doi.org/10.1186/s13059-020-01982-9.

4. Van Nostrand, E.L., Freese, P., Pratt, G.A., Wang, X., Wei, X., Xiao, R., Blue, S.M., Chen, J.Y., Cody, N.A.L., Dominguez, D., et al. (2020). A large-scale binding and functional map of human RNA-binding proteins. Nature *583*, 711–719. https://doi.org/10.1038/s41586-020-2077-3.

5. Fazal, F.M., Han, S., Parker, K.R., Kaewsapsak, P., Xu, J., Boettiger, A.N., Chang, H.Y., and Ting, A.Y. (2019). Atlas of Subcellular RNA Localization Revealed by APEX-Seq. Cell *178*, 473–490.e26. https://doi.org/10.1016/j.cell.2019.05.027.

6. Barutcu, A.R., Wu, M., Braunschweig, U., Dyakov, B.J.A., Luo, Z., Turner, K.M., Durbic, T., Lin, Z.Y., Weatheritt, R.J., Maass, P.G., et al. (2022). Systematic mapping of nuclear domain-associated transcripts reveals speckles and lamina as hubs of functionally distinct retained introns. Mol. Cell *82*, 1035–1052.e9. https://doi.org/10.1016/j.molcel.2021.12.010.

7. Tammer, L., Hameiri, O., Keydar, I., Roy, V.R., Ashkenazy-Titelman, A., Custódio, N., Sason, I., Shayevitch, R., Rodríguez-Vaello, V., Rino, J., et al. (2022). Gene architecture directs splicing outcome in separate nuclear spatial regions. Mol. Cell *82*, 1021–1034.e8. https://doi.org/10.1016/j.molcel.2022.02.001.

8. Tauber, D., Tauber, G., Khong, A., Van Treeck, B., Pelletier, J., and Parker, R. (2020). Modulation of RNA Condensation by the DEAD-Box Protein eIF4A. Cell *180*, 411–426.e16. https://doi.org/10.1016/j.cell.2019.12.031.

9. Banani, S.F., Lee, H.O., Hyman, A.A., and Rosen, M.K. (2017). Biomolecular condensates: organizers of cellular biochemistry. Nat. Rev. Mol. Cell Biol. *18*, 285–298. https://doi.org/10.1038/nrm.2017.7.

10. Quinodoz, S.A., Jachowicz, J.W., Bhat, P., Ollikainen, N., Banerjee, A.K., Goronzy, I.N., Blanco, M.R., Chovanec, P., Chow, A., Markaki, Y., et al. (2021). RNA promotes the formation of spatial compartments in the nucleus. Cell *184*, 5775–5790.e30. https://doi.org/10.1016/j.cell.2021.10.014.

11. Zbinden, A., Pérez-Berlanga, M., De Rossi, P., and Polymenidou, M. (2020). Phase Separation and Neurodegenerative Diseases: A Disturbance in the Force. Dev. Cell *55*, 45–68. https://doi.org/10.1016/j.devcel.2020.09.014.

12. Arnold, E.S., Ling, S.C., Huelga, S.C., Lagier-Tourenne, C., Polymenidou, M., Ditsworth, D., Kordasiewicz, H.B., McAlonis-Downes, M., Platoshyn, O., Parone, P.A., et al. (2013). ALS-linked TDP-43 mutations produce aberrant RNA splicing and adult-onset motor neuron disease without aggregation or loss of nuclear TDP-43. Proc. Natl. Acad. Sci. USA *110*, E736–E745. https://doi.org/10.1073/pnas.1222809110.

13. Montes, M., Sanford, B.L., Comiskey, D.F., and Chandler, D.S. (2019). RNA Splicing and Disease: Animal Models to Therapies. Trends Genet. *35*, 68–87. https://doi.org/10.1016/j.tig.2018.10.002.

14. Tillberg, P.W., Chen, F., Piatkevich, K.D., Zhao, Y., Yu, C.C.J., English, B.P., Gao, L., Martorell, A., Suk, H.J., Yoshida, F., et al. (2016). Protein-retention expansion microscopy of cells and tissues labeled using standard fluorescent proteins and antibodies. Nat. Biotechnol. *34*, 987–992. https://doi.org/10.1038/nbt.3625.

15. Asano, S.M., Gao, R., Wassie, A.T., Tillberg, P.W., Chen, F., and Boyden, E.S. (2018). Expansion Microscopy: Protocols for Imaging Proteins and RNA in Cells and Tissues. Curr. Protoc. Cell Biol. *80*, e56. https://doi.org/10.1002/cpcb.56.

16. Nussbacher, J.K., Tabet, R., Yeo, G.W., and Lagier-Tourenne, C. (2019). Disruption of RNA Metabolism in Neurological Diseases and Emerging Therapeutic Interventions. Neuron *102*, 294–320. https://doi.org/10.1016/j.neuron.2019.03.014.

17. Zhang, Y., Cao, X., Gao, Z., Ma, X., Wang, Q., Xu, X., Cai, X., Zhang, Y., Zhang, Z., Wei, G., et al. (2023). MATR3-antisense LINE1 RNA meshwork scaffolds higher-order chromatin organization. EMBO Rep. *24*, e57550. https://doi.org/10.15252/embr.202357550.

18. Takakuwa, H., Yamazaki, T., Souquere, S., Adachi, S., Yoshino, H., Fujiwara, N., Yamamoto, T., Natsume, T., Nakagawa, S., Pierron, G., et al. (2023). Shell protein composition specified by the lncRNA NEAT1 domains dictates the formation of paraspeckles as distinct membraneless organelles. Nat. Cell Biol. *25*, 1664–1675. https://doi.org/10.1038/s41556-023-01254-1.

19. Platani, M., Goldberg, I., Swedlow, J.R., and Lamond, A.I. (2000). In vivo analysis of Cajal body movement, separation, and joining in live human cells. J. Cell Biol. *151*, 1561–1574. https://doi.org/10.1083/jcb.151.7.1561.

20. Ogg, S.C., and Lamond, A.I. (2002). Cajal bodies and coilin–moving towards function. J. Cell Biol. *159*, 17–21. https://doi.org/10.1083/jcb.200206111.

21. Foley, J.F., Dietrich, D.R., Swenberg, J.A., and Maronpot, R.R. (1991). Detection and Evaluation of Proliferating Cell Nuclear Antigen (Pcna) in Rat Tissue by an Improved Immunohistochemical Procedure. J. Histotechnol. *14*, 237–241. https://doi.org/10.1179/his.1991.14.4.237.

22. Allaway, K.C., Gabitto, M.I., Wapinski, O., Saldi, G., Wang, C.Y., Bandler, R.C., Wu, S.J., Bonneau, R., and Fishell, G. (2021). Genetic and epigenetic coordination of cortical interneuron development. Nature *597*, 693–697. https://doi.org/10.1038/s41586-021-03933-1.

23. Thickman, K.R., Swenson, M.C., Kabogo, J.M., Gryczynski, Z., and Kielkopf, C.L. (2006). Multiple U2AF65 binding sites within SF3b155: thermodynamic and spectroscopic characterization of protein-protein interactions among pre-mRNA splicing factors. J. Mol. Biol. *356*, 664–683. https://doi.org/10.1016/j.jmb.2005.11.067.

24. Tari, M., Manceau, V., de Matha Salone, J., Kobayashi, A., Pastré, D., and Maucuer, A. (2019). U2 AF 65 assemblies drive sequence-specific splice site recognition. EMBO Rep. *20*, e47604. https://doi.org/10.15252/embr.201847604.

25. Malik, A.M., and Barmada, S.J. (2021). Matrin 3 in neuromuscular disease: physiology and pathophysiology. JCI Insight *6*, e143948. https://doi.org/10.1172/jci.insight.143948.

26. Markmiller, S., Soltanieh, S., Server, K.L., Mak, R., Jin, W., Fang, M.Y., Luo, E.C., Krach, F., Yang, D., Sen, A., et al. (2018). Context-Dependent and Disease-Specific Diversity in Protein Interactions within Stress Granules. Cell *172*, 590–604.e13. https://doi.org/10.1016/j.cell.2017.12.032.

27. Naganuma, T., Nakagawa, S., Tanigawa, A., Sasaki, Y.F., Goshima, N., and Hirose, T. (2012). Alternative 3′-end processing of long noncoding RNA initiates construction of nuclear paraspeckles. EMBO J. *31*, 4020–4034. https://doi.org/10.1038/emboj.2012.251.

28. Dopie, J., Sweredoski, M.J., Moradian, A., and Belmont, A.S. (2020). Tyramide signal amplification mass spectrometry (TSA-MS) ratio identifies nuclear speckle proteins. J. Cell Biol. *219*, e201910207. https://doi.org/10.1083/jcb.201910207.

29. Andersen, J.S., Lyon, C.E., Fox, A.H., Leung, A.K.L., Lam, Y.W., Steen, H., Mann, M., and Lamond, A.I. (2002). Directed proteomic analysis of the human nucleolus. Curr. Biol. *12*, 1–11. https://doi.org/10.1016/s0960-9822(01)00650-9.

30. Graindorge, A., Pinheiro, I., Nawrocka, A., Mallory, A.C., Tsvetkov, P., Gil, N., Carolis, C., Buchholz, F., Ulitsky, I., Heard, E., et al. (2019). In-cell identification and measurement of RNA-protein interactions. Nat. Commun. *10*, 5317. https://doi.org/10.1038/s41467-019-13235-w.

31. Necci, M., Piovesan, D.; CAID Predictors; DisProt Curators, and Tosatto, S.C.E. (2021). Critical assessment of protein intrinsic disorder prediction. Nat. Methods *18*, 472–481. https://doi.org/10.1038/s41592-021-01117-3.

32. Sequeira, J.C., Rocha, M., Alves, M.M., and Salvador, A.F. (2022). UPIMAPI, reCOGnizer and KEGGCharter: Bioinformatics tools for functional annotation and visualization of (meta)-omics datasets. Comput. Struct. Biotechnol. J. *20*, 1798–1810. https://doi.org/10.1016/j.csbj.2022.03.042.

33. Mészáros, B., Erdős, G., and Dosztányi, Z. (2018). IUPred2A: context-dependent prediction of protein disorder as a function of redox state and protein binding. Nucleic Acids Res. *46*, W329–W337. https://doi.org/10.1093/nar/gky384.

34. Ray, D., Kazan, H., Cook, K.B., Weirauch, M.T., Najafabadi, H.S., Li, X., Gueroussov, S., Albu, M., Zheng, H., Yang, A., et al. (2013). A compendium of RNA-binding motifs for decoding gene regulation. Nature *499*, 172–177. https://doi.org/10.1038/nature12311.

35. Seal, R.L., Braschi, B., Gray, K., Jones, T.E.M., Tweedie, S., Haim-Vilmovsky, L., and Bruford, E.A. (2023). Genenames.org: the HGNC resources in 2023. Nucleic Acids Res. *51*, D1003–D1009. https://doi.org/10.1093/nar/gkac888.

36. Coelho, M.B., Attig, J., Bellora, N., König, J., Hallegger, M., Kayikci, M., Eyras, E., Ule, J., and Smith, C.W.J. (2015). Nuclear matrix protein Matrin3 regulates alternative splicing and forms overlapping regulatory networks with PTB. EMBO J. *34*, 653–668. https://doi.org/10.15252/embj.201489852.

37. Attig, J., Agostini, F., Gooding, C., Chakrabarti, A.M., Singh, A., Haberman, N., Zagalak, J.A., Emmett, W., Smith, C.W.J., Luscombe, N.M., et al. (2018). Heteromeric RNP Assembly at LINEs Controls Lineage-Specific RNA Processing. Cell *174*, 1067–1081.e17. https://doi.org/10.1016/j.cell.2018.07.001.

38. Mercer, T.R., Clark, M.B., Andersen, S.B., Brunck, M.E., Haerty, W., Crawford, J., Taft, R.J., Nielsen, L.K., Dinger, M.E., and Mattick, J.S. (2015). Genome-wide discovery of human splicing branchpoints. Genome Res. *25*, 290–303. https://doi.org/10.1101/gr.182899.114.

39. Zhang, Y., Liu, T., Meyer, C.A., Eeckhoute, J., Johnson, D.S., Bernstein, B.E., Nusbaum, C., Myers, R.M., Brown, M., Li, W., et al. (2008). Model-based analysis of ChIP-Seq (MACS). Genome Biol. *9*, R137. https://doi.org/10.1186/gb-2008-9-9-r137.

40. Masuda, A., Kawachi, T., Takeda, J.I., Ohkawara, B., Ito, M., and Ohno, K. (2020). tRIP-seq reveals repression of premature polyadenylation by co-transcriptional FUS-U1 snRNP assembly. EMBO Rep. *21*, e49890. https://doi.org/10.15252/embr.201949890.

41. Sun, S., Ling, S.C., Qiu, J., Albuquerque, C.P., Zhou, Y., Tokunaga, S., Li, H., Qiu, H., Bui, A., Yeo, G.W., et al. (2015). ALS-causative mutations in FUS/TLS confer gain and loss of function by altered association with SMN and U1-snRNP. Nat. Commun. *6*, 6171. https://doi.org/10.1038/ncomms7171.

42. Jutzi, D., Campagne, S., Schmidt, R., Reber, S., Mechtersheimer, J., Gypas, F., Schweingruber, C., Colombo, M., von Schroetter, C., Loughlin, F.E., et al. (2020). Aberrant interaction of FUS with the U1 snRNA provides a molecular mechanism of FUS induced amyotrophic lateral sclerosis. Nat. Commun. *11*, 6341. https://doi.org/10.1038/s41467-020-20191-3.

43. Bracha, D., Walls, M.T., Wei, M.T., Zhu, L., Kurian, M., Avalos, J.L., Toettcher, J.E., and Brangwynne, C.P. (2019). Mapping Local and Global Liquid Phase Behavior in Living Cells Using Photo-Oligomerizable Seeds. Cell *176*, 407. https://doi.org/10.1016/j.cell.2018.12.026.

44. Malik, A.M., Miguez, R.A., Li, X., Ho, Y.S., Feldman, E.L., and Barmada, S.J. (2018). Matrin 3-dependent neurotoxicity is modified by nucleic acid binding and nucleocytoplasmic localization. eLife *7*, e35977. https://doi.org/10.7554/eLife.35977.

45. Ma, W., Zhen, G., Xie, W., and Mayr, C. (2021). In vivo reconstitution finds multivalent RNA-RNA interactions as drivers of mesh-like condensates. eLife *10*, e64252. https://doi.org/10.7554/eLife.64252.

46. He, D.C., Nickerson, J.A., and Penman, S. (1990). Core filaments of the nuclear matrix. J. Cell Biol. *110*, 569–580. https://doi.org/10.1083/jcb.110.3.569.

47. Albrethsen, J., Knol, J.C., and Jimenez, C.R. (2009). Unravelling the nuclear matrix proteome. J. Proteomics *72*, 71–81. https://doi.org/10.1016/j.jprot.2008.09.005.

48. Ferrie, J.J., Karr, J.P., Tjian, R., and Darzacq, X. (2022). "Structure"-function relationships in eukaryotic transcription factors: the role of intrinsically disordered regions in gene regulation. Mol. Cell *82*, 3970–3984. https://doi.org/10.1016/j.molcel.2022.09.021.

49. Lyons, H., Veettil, R.T., Pradhan, P., Fornero, C., De La Cruz, N., Ito, K., Eppert, M., Roeder, R.G., and Sabari, B.R. (2023). Functional partitioning of transcriptional regulators by patterned charge blocks. Cell *186*, 327–345.e28. https://doi.org/10.1016/j.cell.2022.12.013.

50. Oksuz, O., Henninger, J.E., Warneford-Thomson, R., Zheng, M.M., Erb, H., Vancura, A., Overholt, K.J., Hawken, S.W., Banani, S.F., Lauman, R., et al. (2023). Transcription factors interact with RNA to regulate genes. Mol. Cell *83*, 2449–2463.e13. https://doi.org/10.1016/j.molcel.2023.06.012.

51. Martin, M. (2011). Cutadapt Removes Adapter Sequences From High-Throughput Sequencing Reads. EMBnet J. *17*, 10–12. https://doi.org/10.14806/ej.17.1.200.

52. Dobin, A., Davis, C.A., Schlesinger, F., Drenkow, J., Zaleski, C., Jha, S., Batut, P., Chaisson, M., and Gingeras, T.R. (2013). STAR: ultrafast universal RNA-seq aligner. Bioinformatics *29*, 15–21. https://doi.org/10.1093/bioinformatics/bts635.

53. Li, B., and Dewey, C.N. (2011). RSEM: accurate transcript quantification from RNA-Seq data with or without a reference genome. BMC Bioinformatics *12*, 323. https://doi.org/10.1186/1471-2105-12-323.

54. Love, M.I., Huber, W., and Anders, S. (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. Genome Biol. *15*, 550. https://doi.org/10.1186/s13059-014-0550-8.

55. Lin, K.T., and Krainer, A.R. (2019). PSI-Sigma: a comprehensive splicing-detection method for short-read and long-read RNA-seq analysis. Bioinformatics *35*, 5048–5054. https://doi.org/10.1093/bioinformatics/btz438.

56. Bailey, T.L., Johnson, J., Grant, C.E., and Noble, W.S. (2015). The MEME suite. Nucleic Acids Res. *43*, W39–W49. https://doi.org/10.1093/nar/gkv416.

57. Erdős, G., Pajkos, M., and Dosztányi, Z. (2021). IUPred3: prediction of protein disorder enhanced with unambiguous experimental annotation and visualization of evolutionary conservation. Nucleic Acids Res. *49*, W297–W303. https://doi.org/10.1093/nar/gkab408.

58. Huang da, W., Sherman, B.T., and Lempicki, R.A. (2009). Systematic and integrative analysis of large gene lists using DAVID bioinformatics resources. Nat. Protoc. *4*, 44–57. https://doi.org/10.1038/nprot.2008.211.

59. Sherman, B.T., Hao, M., Qiu, J., Jiao, X., Baseler, M.W., Lane, H.C., Imamichi, T., and Chang, W. (2022). DAVID: a web server for functional enrichment analysis and functional annotation of gene lists (2021 update). Nucleic Acids Res. *50*, W216–W221. https://doi.org/10.1093/nar/gkac194.

60. Szklarczyk, D., Gable, A.L., Lyon, D., Junge, A., Wyder, S., Huerta-Cepas, J., Simonovic, M., Doncheva, N.T., Morris, J.H., Bork, P., et al. (2019). STRING v11: protein-protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets. Nucleic Acids Res. *47*, D607–D613. https://doi.org/10.1093/nar/gky1131.

61. Heinz, S., Benner, C., Spann, N., Bertolino, E., Lin, Y.C., Laslo, P., Cheng, J.X., Murre, C., Singh, H., and Glass, C.K. (2010). Simple combinations of lineage-determining transcription factors prime cis-regulatory elements required for macrophage and B cell identities. Mol. Cell *38*, 576–589. https://doi.org/10.1016/j.molcel.2010.05.004.

62. Ramírez, F., Ryan, D.P., Grüning, B., Bhardwaj, V., Kilpert, F., Richter, A.S., Heyne, S., Dündar, F., and Manke, T. (2016). deepTools2: a next generation web server for deep-sequencing data analysis. Nucleic Acids Res. *44*, W160–W165. https://doi.org/10.1093/nar/gkw257.

63. Li, H., Handsaker, B., Wysoker, A., Fennell, T., Ruan, J., Homer, N., Marth, G., Abecasis, G., Durbin, R., and Genome Project Data Processing, S. (2009). The Sequence Alignment/Map format and SAMtools. Bioinformatics *25*, 2078–2079. https://doi.org/10.1093/bioinformatics/btp352.

64. Schindelin, J., Arganda-Carreras, I., Frise, E., Kaynig, V., Longair, M., Pietzsch, T., Preibisch, S., Rueden, C., Saalfeld, S., Schmid, B., et al. (2012). Fiji: an open-source platform for biological-image analysis. Nat. Methods *9*, 676–682. https://doi.org/10.1038/nmeth.2019.

65. Arzt, M., Deschamps, J., Schmied, C., Pietzsch, T., Schmidt, D., Tomancak, P., Haase, R., and Jug, F. (2022). LABKIT: Labeling and Segmentation Toolkit for Big Image Data. Front. Comput. Sci. *4*, 777728. https://doi.org/10.3389/fcomp.2022.777728.

66. Lucy, L.B. (1974). An iterative technique for the rectification of observed distributions. Astron. J. *79*, 745. https://doi.org/10.1086/111605.

67. Richardson, W.H. (1972). Bayesian-Based Iterative Method of Image Restoration. J. Opt. Soc. Am. *62*, 55–59. https://doi.org/10.1364/JOSA.62.000055.

68. Fu, Y., Masuda, A., Ito, M., Shinmi, J., and Ohno, K. (2011). AG-dependent 3′-splice sites are predisposed to aberrant splicing due to a mutation at the first nucleotide of an exon. Nucleic Acids Res. *39*, 4396–4404. https://doi.org/10.1093/nar/gkr026.

69. Mizushima, S., and Nagata, S. (1990). pEF-BOS, a powerful mammalian expression vector. Nucleic Acids Res. *18*, 5322. https://doi.org/10.1093/nar/18.17.5322.

70. Kato, M., Han, T.W., Xie, S., Shi, K., Du, X., Wu, L.C., Mirzaei, H., Goldsmith, E.J., Longgood, J., Pei, J., et al. (2012). Cell-free Formation of RNA Granules: Low Complexity Sequence Domains Form Dynamic Fibers within Hydrogels. Cell *149*, 753–767. https://doi.org/10.1016/j.cell.2012.04.017.

71. Kawachi, T., Masuda, A., Yamashita, Y., Takeda, J.I., Ohkawara, B., Ito, M., and Ohno, K. (2021). Regulated splicing of large exons is linked to phase-separation of vertebrate transcription factors. EMBO J. *40*, e107485. https://doi.org/10.15252/embj.2020107485.

72. Cunningham, F., Allen, J.E., Allen, J., Alvarez-Jarreta, J., Amode, M.R., Armean, I.M., Austine-Orimoloye, O., Azov, A.G., Barnes, I., Bennett, R., et al. (2022). Ensembl 2022. Nucleic Acids Res. *50*, D988–D995. https://doi.org/10.1093/nar/gkab1049.

73. Kyte, J., and Doolittle, R.F. (1982). A simple method for displaying the hydropathic character of a protein. J. Mol. Biol. *157*, 105–132. https://doi.org/10.1016/0022-2836(82)90515-0.

74. Letunic, I., Khedkar, S., and Bork, P. (2021). SMART: recent updates, new developments and status in 2020. Nucleic Acids Res. *49*, D458–D460. https://doi.org/10.1093/nar/gkaa937.

75. Ogata, H., Goto, S., Fujibuchi, W., and Kanehisa, M. (1998). Computation with the KEGG pathway database. Biosystems. *47*, 119–128. https://doi.org/10.1016/s0303-2647(98)00017-3.

76. Masuda, A., Takeda, J., Okuno, T., Okamoto, T., Ohkawara, B., Ito, M., Ishigaki, S., Sobue, G., and Ohno, K. (2015). Position-specific binding of FUS to nascent RNA regulates mRNA length. Genes Dev. *29*, 1045–1057. https://doi.org/10.1101/gad.255737.114.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Mouse monoclonal anti-ACTB	Santa Cruz Biotechnology	Cat# sc-47778, RRID:AB_626632
Mouse monoclonal anti-BrdU	Santa Cruz Biotechnology	Cat# sc-32323, RRID:AB_626766
Mouse monoclonal anti-CELF1	Santa Cruz Biotechnology	Cat# sc-20003, RRID:AB_627319
Rabbit monoclonal anti-Coilin	Cell signaling Technology	Cat# 14168, RRID:AB_2798410
Mouse monoclonal anti-EWSR1	Santa Cruz Biotechnology	Cat# sc-28327, RRID:AB_675526
Mouse monoclonal anti-FLAG (M2)	Sigma-Aldrich	Cat# P2983, RRID:AB_439685
Mouse monoclonal anti-FUS	Santa Cruz Biotechnology	Cat# sc-47711, RRID:AB_2105208
Rabbit monoclonal anti-FUS	Bethyl Laboratories	Cat# A300-302A, RRID:AB_309445
Mouse monoclonal anti-GFP	Roche	Cat# 11814460001, RRID:AB_390913
Mouse monoclonal anti-hnRNP A1	Santa Cruz Biotechnology	Cat# sc-32301, RRID:AB_627729
Mouse monoclonal anti-m3G/TMG (5′ cap)	Sigma-Aldrich	Cat# MABE302, RRID:AB_2941969
Rabbit monoclonal anti-MATR3	Abcam	Cat# ab151714, RRID:AB_2491618
Mouse monoclonal anti-NeuN	Sigma-Aldrich	Cat# MAB377, RRID:AB_2298772
Mouse monoclonal anti-PCNA	Santa Cruz Biotechnology	Cat# sc-56, RRID:AB_628110
Rat monoclonal anti-pSer2 RNAP II	Sigma-Aldrich	Cat# 04-1571, RRID:AB_10627998
Mouse monoclonal anti-PTBP1	Thermo Fisher Scientific	Cat# 32-4800, RRID:AB_2533082
Rabbit polyclonal anti-PTBP1	Proteintech	Cat# 12582-1-AP, RRID:AB_2176402
Mouse monoclonal anti-SF3B1	Santa Cruz Biotechnology	Cat# sc-514655, RRID:AB_2941964
Mouse monoclonal anti-SNRPA1	Santa Cruz Biotechnology	Cat# sc-393804, RRID:AB_2941967
Rabbit polyclonal anti-TAF15	Bethyl Laboratories	Cat# A300-309A, RRID:AB_309448
Mouse monoclonal anti-SON	Santa Cruz Biotechnology	Cat# sc-398508, RRID:AB_2868584
Rabbit polyclonal anti-TDP43	Bethyl Laboratories	Cat# A303-223A, RRID:AB_10973681
Mouse monoclonal anti-U1-70K	Synaptic Systems	Cat# 203 011, RRID:AB_887903
Rabbit polyclonal anti-U1A	Thermo Fisher Scientific	Cat# PA5-27474, RRID:AB_2544950
Rat monoclonal anti-U1C	BioAcademia	Cat# 70-400, RRID:AB_10550948
Mouse monoclonal anti-U2AF2	Santa Cruz Biotechnology	Cat# sc-53942, RRID:AB_831787
Recombinant DNA		
pEF-Bos-MATR3-EGFP (MATR3-EGFP)	This paper	N/A
pEF-Bos-MATR3-mCherry (MATR3-mCherry)	This paper	N/A
pEF-Bos-MATR3-EGFP-ΔIDR	This paper	N/A
pEF-Bos-MATR3-EGFP-PTBP1-IDR	This paper	N/A
pEF-Bos-MATR3-EGFP-SF3B1-IDR	This paper	N/A
p3xFLAG-CMV10-MATR3 (WT)	This paper	N/A
p3xFLAG-CMV10-MATR3-ΔRRM1	This paper	N/A
p3xFLAG-CMV10-MATR3-ΔRRM2	This paper	N/A
p3xFLAG-CMV10-MATR3-ΔIDR	This paper	N/A
p3xFLAG-CMV10-AcGFP-U2AF2 (U2_WT)	This paper	N/A
p3xFLAG-CMV10-mCherry-U2AF2 (U2_WT-mCherry)	This paper	N/A
p3xFLAG-CMV10-mCherry-U2AF2ΔIDR (U2_ ΔIDR)	This paper	N/A
p3xFLAG-CMV10-mCherry-U2AF2-MATR3_A IDR (U2_MA3A IDR)	This paper	N/A

(Continued on next page)

Continued		
REAGENT or RESOURCE	SOURCE	IDENTIFIER
p3xFLAG-CMV10-mCherry-U2AF2-SF3B1 IDR (U2_SF3B1 IDR)	This paper	N/A
pHR-SFFVp-MATR3::mCherry::SspB	This paper	N/A
pHR-SFFVp-NLS-iLID::EGFP::FTH1	Addgene	#122147
pEGFP-N1	Clontech	6085-1
pEGFP-N1-PTBP1 (PTBP1-EGFP)	This paper	N/A
pRSETA-AcGFP-MATR3_IDR	This paper	N/A
pRSETA-AcGFP-MATR3_IDR_A	This paper	N/A
pRSETA-AcGFP-MATR3_IDR_B	This paper	N/A
pET28-AcGFP-U2AF2	This paper	N/A
pET28-mCherry-U2AF2	This paper	N/A
pET28-mCherry-U2AF2-ΔIDR	This paper	N/A
pET28-AcGFP-PTBP1	This paper	N/A
pET28-mCherry-PTBP1	This paper	N/A
pET28-AcGFP-PTBP1-ΔIDR	This paper	N/A
pET28-mCherry-PTBP1-ΔIDR	This paper	N/A
pET28-AcGFP-CELF1	This paper	N/A
pET28-mCherry-CELF1	This paper	N/A
pET28-AcGFP/-U170K	This paper	N/A
pET28-mCherry-U170K	This paper	N/A
pET28-mCherry-CELF1-WT	This paper	N/A
pET28-mCherry-CELF1-ΔIDR	This paper	N/A
Chemicals, peptides, and recombinant proteins		
Acryloyl-X SE (AcX)	Thermo Fisher Scientific	Cat# A20770
Sodium acrylate	Sigma-Aldrich	Cat# 408220
N, N′-methylene bisacrylamide	Sigma-Aldrich	Cat# M7279
DABCO	Sigma-Aldrich	Cat# 290734
SeaPlaque agarose	LONZA	Cat# 50101
5,6-dichlorobenzimidazole riboside (DRB)	Sigma Aldrich	Cat# D1916
Critical commercial assays		
Lipofectamine 3000 Trancfection Reagent	Thermo Fisher Scientific	Cat# L3000015
Lipofectamine RNAiMAX Trancfection Reagent	Thermo Fisher Scientific	Cat# 13778150
Lipofectamine CRISPRMAX Cas9 Trancfection Reagent	Thermo Fisher Scientific	Cat# CMAX00008
FuGENE 6	Promega	Cat# E269A
Proteinase K	New England Biolabs Inc.	Cat# P8107S
Deposited data		
CLIP-seq of the endogenous MATR3 (N2A)	This paper	DDBJ: DRR519925
CLIP-seq of the endogenous MATR3 (N2A, replicate 1)	This paper	DDBJ: DRR519926
CLIP-seq of the endogenous MATR3 (N2A, replicate 2)	This paper	DDBJ: DRR519927
CLIP-seq of the endogenous MATR3 (Endo)	This paper	DDBJ: DRR494386
CLIP-seq of MATR3 (WT)	This paper	DDBJ: DRR494387
CLIP-seq of MATR3 (WT, replicate)	This paper	DDBJ: DRR519006
CLIP-seq of MATR3 (ΔRRM1)	This paper	DDBJ: DRR494388
CLIP-seq of MATR3 (ΔRRM2)	This paper	DDBJ: DRR494389
CLIP-seq of MATR3 (ΔIDR)	This paper	DDBJ: DRR494390
CLIP-seq of MATR3 (ΔIDR, replicate)	This paper	DDBJ: DRR519007
CLIP-seq of U2AF2 (U2_WT)	This paper	DDBJ: DRR519008
CLIP-seq of U2AF2 (U2_ΔIDR)	This paper	DDBJ: DRR519009
CLIP-seq of PTBP1 using control siRNA-treated cells	This paper	DDBJ: DRR494396

(*Continued on next page*)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
CLIP-seq of PTBP1 using untreated cells	This paper	DDBJ: DRR494399
CLIP-seq of PTBP1 using Matr3-KD cells, treated with siMatr3-1	This paper	DDBJ: DRR494397
CLIP-seq of PTBP1 using Matr3-KD cells, treated with siMatr3-2	This paper	DDBJ: DRR494398
CLIP-seq of SF3B1 using control siRNA-treated cells	This paper	DDBJ: DRR519010
CLIP-seq of SF3B1 using untreated cells	This paper	DDBJ: DRR494395
CLIP-seq of SF3B1 using Matr3-KD cells, treated with siMatr3-1	This paper	DDBJ: DRR494393
CLIP-seq of SF3B1 using Matr3-KD cells, treated with siMatr3-2	This paper	DDBJ: DRR494394
CLIP-seq of U1C using control cells	Masuda et al.[40]	DDBJ: DRR171207
CLIP-seq of U1C using Fus-KD cells	Masuda et al.[40]	DDBJ: DRR171208
CLIP-seq of U1-70k using control cells	Masuda et al.[40]	DDBJ: DRR171206
CLIP-seq of U1-70K using Fus-KD cells	Masuda et al.[40]	DDBJ: DRR494391
RNA-seq of cells, where condensation of MATR3-mCherry-SspB was light-induced. Biological replicate 1	This paper	DDBJ: DRR494400
RNA-seq of cells, where condensation of MATR3-mCherry-SspB was light-induced. Biological replicate 2	This paper	DDBJ: DRR494401
RNA-seq of non-irradiated control cells	This paper	DDBJ: DRR494402
RNA-seq of irradiated control cells, expressing EGFP alone	This paper	DDBJ: DRR494403
RNA-seq of HEK293 cells treated with control siRNA	This paper	DDBJ: DRR519013
RNA-seq of HEK293 cells treated with control siRNA (replicate)	This paper	DDBJ: DRR519014
RNA-seq of HEK293 cells treated with siMATR3-1	This paper	DDBJ: DRR519011
RNA-seq of HEK293 cells treated with siMATR3-2	This paper	DDBJ: DRR519012
Unprocessed raw data in Mendeley Data	This paper	https://doi.org/10.17632/72d9dyrf7r.1
Experimental models: Cell lines		
Mouse: neuroblastoma cell line, N2A	RIKEN BioResource Research Center	RCB0987
Human: HEK 293	In house	N/A
Bacterial and virus strains		
BL21 (DE3)	NEB	Cat# C2527H
Oligonucleotides (PCR primers)		
5′-TAATACGACTCACTATAGGG TTGC AGTCTATATTTAACAT GTT TTAGAGCTAGAAATA-3′	Fasmac	T7-sgRNA-F
5′-AAAAGCACCGACTCGGTGCC-3′	Fasmac	T7-sgRNA-R
5′-TGCCAGCATCTCTTGGAAGG-3′	Fasmac	Matr3-F
5′-GTATGGTGGCTCCCGTGTAG-3′	Fasmac	Matr3-R
Software and algorithms		
Cutadapt	Martin[51]	cutadapt v3.7
STAR	Dobin et al.[52]	STAR_2.7.9a
RSEM	Li and Dewey[53]	RSEM v1.3.3
DeSEQ2	Love et al.[54]	DeSEQ2 v1.30.1

(*Continued on next page*)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
PSI-SIGMA	Lin and Krainer[55]	https://github.com/wososa/PSI-Sigma
MACS	Zhang et al.[39]	MACS2 2.1.4
MEME	Bailey et al.[56]	MEME-SUITE (v5.5.1)
reCOGnizer	Sequeira et al.[32]	reCOGnizer 1.8.1
IUPred3	Erdos et al.[57]	https://iupred3.elte.hu/download_new
DAVID Bioinformatics Resources	Huang da et al.,[58] Sherman et al.[59]	https://david.ncifcrf.gov/tools.jsp
STRING	Szklarczyk et al.[60]	https://string-db.org/
HOMER	Heinz et al.[61]	HOMER v4.11
deepTools	Ramirez et al.[62]	deeptools 3.5.1
samtools	Li et al.[63]	Samtools 1.14
R	https://www.r-project.org	R version 4.2.2
JMP	SAS Institute Inc	JMP, version 16.1.0
Fiji	Schindelin et al.[64]	Fiji 2.13.1
Labkit	Arzt et al.[65]	Labkit 0.3.11
Other		
siRNA against *Matr3* (siMatr3-1), 5′- GGGAAAAUAAAGAAUUAUA-3′	Fasmac	siMatr3-1
siRNA against *Matr3* (siMatr3-2), 5′-GGAUGAUUGGGAAGAAAAA-3′	Fasmac	siMatr3-2
siRNA against *MATR3* (siMATR3-1), 5′- CCAGAAGUAUAAAAGAAUA -3′	Fasmac	siMATR3-1
siRNA against *MATR3* (siMATR3-2), 5′- GUUGGAAAAUGGAAUUAAA -3′	Fasmac	siMATR-2
AllStar Negative Control siRNA	Qiagen	siCont, Cat# 1027281
No. 1.5H glass coverslips	Marienfeld	Cat# 107032
No. 1 glass coverslips	Matsunami	Cat# C012001
No. 1S/1.5 glass bottom dish	Matsunami	Cat# D11130H
Blue-Light Transilluminator	OptoCode	Cat# LEDB-SBOXHP

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Akio Masuda (masuda.akio.h1@f.mail.nagoya-u.ac.jp).

Materials availability

All unique/stable reagents generated in this study are available from the lead contact with a completed Materials Transfer Agreement.

Data and code availability

- RNA-seq and CLIP-seq data have been deposited at DDBJ and are publicly available as of the date of publication. The accession numbers for the RNA-seq data are DDBJ: DRA016792, and DRA017583. Those for the CLIP-seq data are DDBJ: DRA016789, DRA016791, DRA016790, DRA008142, DRA017580, DRA017581, DRA017582, and DRA017630. The unprocessed image files have been deposited on Mendeley Data: https://doi.org/10.17632/72d9dyrf7r.1 and are publicly available as of the date of publication. Accession numbers and DOI are listed in the key resources table.
- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Human and mouse sections

Fresh frozen sections of normal human tissues were purchased from BioChain Institute, Inc. (Newark, CA) and Zyagen (San Diego, CA). Fresh frozen sections of mouse embryo (embryonic day 18) were purchased from Zyagen.

Cell lines and cell culture

N2A mouse neuroblastoma cells were grown in MEM with 10% fetal bovine serum at 37°C in 5% CO_2. HEK293 cells were grown in DMEM with 10% fetal bovine serum.

Generation of the N2A cell line carrying an S85C mutation

To generate the N2A cell line carrying an S85C mutation, we used the CRISPR/Cas9 system in combination with a single-stranded oligonucleotide (ssODN) as a homologous template containing the S85C mutation. ssODN was synthesized by Fasmac. The CRISPR/Cas9 target sequence was 5′- TTGCAGTCTATATTTAACAT-3′. The single guide RNA (sgRNAs) was generated by an *in vitro* transcription system as follows. The template for *in vitro* transcription of sgRNA was amplified by PCR using the pX330 vector (Addgene) as template DNA with the primers: forward primer, 5′-TAATACGACTCACTATAGGG-[20-bp sgRNA target sequence]-GTTTTAGAGCTAGAAATA-3′, and reverse primer, 5′-AAAAGCACCGACTCGGTGCC-3′. The underlined sequence in the forward primer indicates the T7 promoter region. The segment of tracer RNA was obtained by PCR using the pX330 vector as a template. The amplified PCR fragment was purified with AMPure XP beads (Beckman Coulter) and was transcribed *in vitro* using RiboMAX Large Scale RNA Production Systems-T7 (Promega). The sgRNA was purified with Quick-RNA MicroPrep Kit (Zymo Research).

N2A cells were transfected with ssODN, sgRNA, and Cas9 enzyme (Integrated DNA Technologies) using Lipofectamine CRISPRMAX (Thermo Fisher Scientific) according to the manufacturer's instructions. Two days after transfection, cells were seeded at 0.5 cells/well and 0.2 cells/well in 96-well plates. After 14 days, all single-cell clones were screened with heteroduplex cleavage assays using the Guide-it Mutation Detection Kit (Takara) with a forward primer, 5′-TGCCAGCATCTCTTGGAAGG-3′, and a reverse primer, 5′-GTATGGTGGCTCCCGTGTAG-3′. The amplicons were also sequenced to confirm the introduction of the S85C mutation.

METHOD DETAILS

Immunostaining

Cells grown on No. 1.5H glass coverslips (Marienfeld) in a 6-well plate, as well as tissue sections on glass slides, were fixed in 4% paraformaldehyde in PBS (Wako) for 10 min at RT. After three washes in PBS for 5 min, specimens were permeabilized with 0.5% Triton X-100 in PBS for 10 min at RT. Following three washes in PBS containing 0.1% Tween 20 (PBST) for 5 min, specimens were blocked with 3% bovine serum albumin (Sigma-Aldrich) for 30 min at RT and incubated with primary antibodies in 3% bovine serum albumin overnight at 4°C. After three washes in PBST, the primary antibodies were recognized by specific secondary antibodies [1:1000 dilution of Goat anti-mouse IgG Alexa Fluor 488 (Thermo Fisher Scientific, A11001), Goat anti-rabbit IgG Alexa Fluor 546 (11010), or Goat anti-rat IgG Alexa Fluor 594 (A11007)] along with 1 mg/ml Hoechst 33258 (Dojindo) in the dark for 1 hour. Cells were washed three times with PBST. Glass slides were mounted onto slides with Prolong Glass Antifade Mountant (Thermo Fisher Scientific). Coverslips were sealed with transparent nail polish and stored at 4°C. Images were acquired with the SpinSR10 spinning disk confocal microscope (Olympus) with a 100× UPLAPO OHR (NA 1.5) oil Objective using a Hamamatsu ORCA-Flash4.0 CMOS camera (Hamamatsu Photonics) and OLYMPUS cellSens Dimension software. Image stacks taken at fixed distances perpendicular to the focal plane (0.1 μm steps) were further processed by the built-in deconvolution function in Olympus cellSens Dimension software with the maximum likelihood estimation algorithm (the Richardson-Lucy, 20 iterations).[66,67]

2D-STORM

N2A cells grown on No. 1.5H glass coverslips (Marienfeld) in a 6-well plate were fixed in 3% EM-grade paraformaldehyde (Electron Microscopy Sciences) and 0.1% EM-grade glutaraldehyde (Electron Microscopy Sciences) in PBS (Wako) for 10 min at RT. Specimens were washed three times with PBS and quenched with 0.1% sodium borohydride (Sigma-Aldrich) in PBS for 7 min. After three washes in PBS for 5 min, specimens were permeabilized with 0.2% Triton X-100 in PBS for 15 min at RT. Following three washes in PBS containing 0.1% Tween 20 (PBST) for 5 min, specimens were blocked with 3% bovine serum albumin (Sigma-Aldrich) for 30 min at RT and incubated with primary antibodies in 3% bovine serum albumin overnight at 4°C. After three washes in PBST, the primary antibody was recognized by the secondary antibody [Goat anti-mouse IgG Alexa Fluor 647 (Thermo Fisher Scientific A21235), 1:1000 dilution] along with 1 mg/ml Hoechst 33258 (Dojindo) in the dark for 1.5 h at RT. Cells were washed five times with PBST and post-fixed with 3% EM-grade paraformaldehyde and 0.1% EM-grade glutaraldehyde in PBS for 10 min at RT. Coverslips were mounted onto slides with freshly prepared oxygen scavenger imaging buffer [50 mM Tris-HCl pH 8.0, 10 mM NaCl, 10% Glucose, 100 mM Cysteamine (Sigma-Aldrich), 0.5 mg/mL Glucose Oxidase (Sigma-Aldrich) and 0.04 mg/mL Catalase (WAKO)] and sealed with transparent nail polish.

The 2D-STORM was carried out using a Nikon Eclipse Ti2-E inverted microscope (Nikon Instruments, Inc.) equipped with the Perfect Focus System, applying an objective-type total internal reflection fluorescence (TIRF) configuration with an oil-immersion objective (CFI SR HP Apo TIRF 100x, NA 1.49, Oil). TIRF angle was adjusted for optimal signal-to-noise ratio. Ten thousand frames were acquired at 10 ms/frame exposure at 100% power of 638 nm laser with 405 nm activation. Emitted light was collected by a Hamamatsu ORCA-Flash4.0 CMOS camera set on a 16-bit scale detection modality.

Imaging reconstruction of 2D-STORM

The localizations of single fluorophore activations were detected using NIS elements (Nikon). The data was processed with a minimum height of the fluorophore point spread function of 50,000 and a minimum photon of 500. A 20 nm radius Gaussian blur (ImageJ/Gaussian blur) was applied.

Expansion microscopy

Sample expansion was carried out as described elsewhere.[14,15] Briefly, cells grown on 12-mm round glass coverslips (No.1 thickness, Matsunami) in a 12-well plate were fluorescently immunostained as described in the ''immunostaining'' section. The cells were then incubated with 0.1 mg/ml Acryloyl-X SE (AcX) (Thermo Fisher Scientific) in PBS for 3 h at RT. After incubation with AcX solution, the cells were washed three times in PBS for 5 min. A drop of 40 μl of the gelling solution containing the monomer solution [1 × PBS, 2 M NaCl, 8.625% (w/w) sodium acrylate (Sigma-Aldrich), 2.5% (w/w) acrylamide (Sigma-Aldrich), 0.15% (w/w) N, N'-methylene bisacrylamide (Sigma-Aldrich), 0.2% (w/v) tetramethylethylenediamine (TEMED) (Sigma-Aldrich), and 0.2% (w/v) ammonium persulfate (APS) (WAKO)] was placed on a parafilm, and the coverslips were inverted over the droplets with the cells facing down, followed by incubation for 1 h at 37°C in a humidified chamber. After gelation, specimens were incubated in the digestion buffer (50 mM Tris pH 8.0, 1 mM EDTA, 0.5% Triton X-100, 0.8 M guanidine HCl) containing 8 U/ml proteinase K (New England Biolabs, Inc.) for 4 h at 37°C. The gel was transferred into a 60 mm dish, and deionized water containing 1 μg/ml Hoechst 33258 was added. The water was exchanged three or four times every 30 min until the gel was fully expanded. Before imaging, the gel was trimmed with a razor blade and incubated in 1% DABCO (Sigma-Aldrich) in deionized water to reduce the fading rate of fluorescein. To prevent the sample from drifting during the imaging, the gel was immobilized with 2% SeaPlaque low-melting-point agarose (LONZA) on No. 1.5H glass coverslips (Marienfeld). Confocal z-stack images (0.1 μm steps) were acquired using the SpinSR10 spinning disk confocal microscope with a 100× UPLAPO OHR oil objective and processed by the built-in deconvolution function in Olympus cellSens Dimension software with the maximum likelihood estimation algorithm (the Richardson-Lucy, 20 iterations).

Construction of mammalian expression vectors and transfection

Coding sequences (CDSs) of human *MATR3* (accession number, BC015031), human *PTBP1* (accession number, BC013694), and human *SF3B1* (accession number, NM_012433) were purchased from GE Healthcare, Open Biosystems, and Kazusa DNA Research Institute, respectively. Human *U2AF2* CDS was obtained as previously described.[68]

To construct the expression vector of EGFP-tagged MATR3 (MATR3-EGFP), PCR-amplified *MATR3* CDS and EGFP CDS were cloned into the pEF-Bos mammalian expression vector[69] at XbaI-BamHI sites and BamHI-MluI sites, respectively. Similarly, the expression vector of mCherry-tagged MATR3 (MATR3-mCherry) was constructed. The expression vector of IDR-deleted MATR3-mCherry (ΔIDR) was generated by replacing the IDR (2-275 aa of NP_001181883) with a KpnI site using the QuikChange site-directed mutagenesis kit (Agilent Technologies). The vectors for the MATR3 mutants, whose IDR was replaced with that of PTBP1 (285-363 aa of NP_002810, PTBP1-IDR) and with that of SF3B1 (2-160 aa of NP_036565, SF3B1-IDR), were generated by inserting the corresponding CDSs into the KpnI site of ΔIDR.

To construct the expression vector of FLAG-tagged MATR3 (WT), PCR-amplified *MATR3* CDS was cloned into the p3xFLAG-CMV-10 mammalian expression vector (Sigma). The expression vectors of FLAG-tagged MATR3 deletion mutants lacking RRM1 (391-490 aa of NP_001181883, ΔRRM1), RRM2 (491-580 aa, ΔRRM2), and IDR (1-275 aa, ΔIDR) were generated using the QuikChange site-directed mutagenesis kit.

To construct the expression vector of FLAG- and AcGFP-tagged U2AF2 (U2_WT), PCR-amplified *U2AF2* CDS and AcGFP were cloned into the p3xFLAG-CMV-10 mammalian expression vector (Sigma). Similarly, the expression vector of FLAG- and mCherry-tagged MATR3 (U2_WT-mCherry) was constructed. The expression vector for the mutant U2AF2 (U2AF2-ΔIDR) lacking the IDR (1-111 aa of NP_009210) was generated by inserting the corresponding segment into an MluI site of U2_WT-mCherry. The vectors for the U2AF2 mutants, whose IDR was replaced with the specific region of MATR3-IDR (MATR3-A: 22-78 aa of NP_001181883, U2_MA3A-IDR) and with the IDR of SF3B1 (2-160 aa of NP_036565, U2_SF3B1-IDR) were generated by inserting the corresponding CDSs into the MluI site of U2_ΔIDR.

pHR-SFFVp-FUSN::mCherry::SspB (#122148) and pHR-SFFVp-NLS-iLID::EGFP::FTH1 (#122147, iLID-EGFP-FTH1) were obtained from Addgene.[43] The expression vector of MATR3 fused to mCherry and SspB (MATR3-mCherry-SspB) was generated by replacing the *FUS* CDS in the pHR-SFFVp-FUSN::mCherry::SspB with *MATR3* CDS using In-Fusion HD cloning Kit (Takara). The pEGFP-N1 vector was purchased from Clontech.

To construct the expression vector of EGFP-tagged PTBP1 (PTBP1-EGFP), PCR-amplified *PTBP1* CDS was cloned into the pEGFP-N1 vector at the HindIII site.

All expression constructs were sequenced to ensure the lack of PCR artifacts. These vectors were introduced into HEK293 cells or N2A cells with Lipofectamine 3000 (Thermo Fisher Scientific) according to the manufacturer's instructions.

BrUTP labeling of nascent RNA in HEK cells

BrUTP (final concentration, 2 mM, Sigma-Aldrich) was mixed with 6 μL of FuGene-6 transfection reagent (Roche) in 100 μL of Opti-MEM (Gibco Life Technology) according to the manufacturer's instructions. Then, HEK293 cells grown on a No. 1.5H glass coverslip were incubated with the mixture for 15 min at 4°C, followed by incubation at 37°C. At 0, 20, 60, and 180 min after starting incubation at

37°C, the cells were gently washed with ice-cold PBS and fixed in 4% paraformaldehyde for 10 min at RT. Immunostaining was performed as described above.

Live cell imaging

N2A cells were cultured in a glass bottom dish (Matsunami) and transfected with the indicated vectors with Lipofectamine 3000 (Thermo Fisher Scientific) according to the manufacturer's instructions. One day after transfection, culture media was changed to phenol-red-free DMEM (Agilent Technologies) with 10% FBS. Images were acquired with the SpinSR10 spinning disk confocal microscope with a 100× UPLAPO OHR (NA 1.5) oil Objective using a Hamamatsu ORCA-Flash4.0 CMOS camera (Hamamatsu Photonics) and OLYMPUS cellSens Dimension software.

Fluorescence recovery after photobleaching assay

Fluorescence recovery after photobleaching (FRAP) assays were performed using the Zeiss LSM880 laser scanning confocal microscope equipped with an alpha Plan-Apochromat 100x/1.46 Oil DIC M27 Elyra objective and Zeiss ZEN software. A circular region of interest was bleached once using 80% laser power with 80 iterations. Fluorescence recovery was quantified using ZEISS ZEN software.

Recombinant proteins

Purified recombinant full-length RBPs (PTBP1 and U2AF2) and fractional RBPs carrying IDRs (FUS,[70] SF3B1,[23] MATR3,[25] CELF1, and U1-70K), which were fused with AcGFP or mCherry, were generated (gray bars in the right panel of Figure 3B), as follows. The bacterial expression vectors for the recombinant proteins used in the *in vitro* droplet formation assays were generated as follows. The CDS of AcGFP was inserted into the pRSET A bacterial expression vector (Thermo Fisher Scientific) at BamHI-XhoI sites (pRSETA-AcGFP). Similarly, the CDSs of AcGFP and mCherry were inserted into the pET28a(+) vector (Novagen) at BamHI-XhoI sites (pET28-AcGFP and pET28-mCherry, respectively). Then, the CDS of MATR3 IDR (2-275 aa of NP_001181883) was inserted into XhoI-EcoRI sites of the pRSETA-AcGFP vector (pRSETA-AcGFP-MATR3_IDR). The CDSs of PTBP1 (NP_002810) and U2AF2 (NP_009210) were inserted into the pET28-AcGFP/mCherry at BamHI-HindIII sites to construct pET28-AcGFP/mCherry-PTBP1 and pET28-AcGFP/mCherry-PTBP1, respectively. The CDS of CELF1 (199-388 aa of NP_006551) and U1-70K (187-437 aa of NP_003080) were inserted into the pET28-AcGFP/mCherry at BamHI-XhoI sites and EcoRI-XhoI sites, respectively (pET28-AcGFP/mCherry-CELF1 and pET28-AcGFP/mCherry-U170K).

The expression vectors for the specific regions of MATR3-IDR (MATR3-A: 22-78 aa and MATR3-B: 123-275 aa of NP_001181883) were generated by the insertion of PCR-amplified corresponding fragments into the pET28-AcGFP-vector. The expression vectors for the mutant PTBP1 (PTBP1-ΔIDR) lacking the IDRs (294-361 aa of NP_002810) were generated by replacing the corresponding segment in pET28-AcGFP/mCherry-PTBP1 with a KpnI site. Similarly, the expression vector for the mutant U2AF2 (U2AF2-ΔIDR) lacking the IDR (1-111 aa of NP_009210) was generated by replacing the corresponding segment in pET28-mCherry-U2AF2 with a MluI site. All expression constructs were sequenced to ensure the lack of PCR artifacts and were transformed into BL21 (DE3) cells (NEB).

The transformed bacteria were grown in LB medium with 100 μg/ml ampicillin. At absorbance OD600 = 0.6, protein expression was induced with 0.1 mM IPTG, and the bacteria were grown for an additional 15 h at 25°C at 150 rpm. After centrifugation, cell pellets were resuspended in 10 ml PBS and sonicated (MySonic, Power 60, 10 cycles of 30-sec burst and 30-sec rest) at 4°C. Histidine-tagged proteins were isolated from the soluble lysate using TALON metal affinity beads (Takara), followed by the elution with 150 mM imidazole. Then, the eluted proteins were concentrated, and the buffer was exchanged to 100 mM Tris-HCl, pH 7.5, and 20% glycerol using Amicon Ultra centrifugal filters (Millipore).

For the IDR of FUS (14-174 aa of NP_004951), the CDS of mCherry-tagged FUS-IDR was cloned into the pGEX6P1 vector (Cytiva) at BamHI-XhoI sites, followed by transformation into BL21 (DE3) cells (NEB).[71] Induction of protein synthesis and sonication of the bacteria were performed as described above. GST-fusion proteins were isolated from the soluble lysate using glutathione-Sepharose 4B beads (Cytiva). Then, the recombinant proteins were cleaved on the beads from GST by incubation with PreScission protease (Cytiva) in PreScission buffer (50 mM Tris-HCl pH 8.0, 100 mM NaCl, 1 mM EDTA, and 1 mM DTT) overnight at 4°C. The cleaved proteins were concentrated, and the buffer was exchanged to 100 mM Tris-HCl, pH 7.5, and 20% glycerol using Amicon Ultra centrifugal filters (Millipore).

For the IDR of SF3B1 (2-511 aa of NP_036565), the CDS of mCherry-tagged SF3B1-IDR was cloned into the pFastBac HT vector at NotI-XhoI sites. The recombinant protein was produced using the Bac-to-Bac Baculovirus Expression System (Thermo Fisher Scientific) according to the manufacturer's instructions and was purified using TALON metal affinity beads (Takara) as described above.

In vitro droplet formation assay

The recombinant proteins were added to solutions at 10 nM final concentration with 75 mM NaCl and 15% PEG-8000 as a crowding agent in Droplet formation buffer (50 mM Tris-HCl pH 7.5, 10% glycerol, and 1 mM DTT). The protein solution was placed on a slide glass, covered with a cover slip, and sealed with nail polish to prevent evaporation. Slides were then imaged with the Nikon A1R laser scanning confocal microscope equipped with a 100× NA 1.45 oil Objective or with the SpinSR10 spinning disk confocal microscope with a 60× UPLXAPO NA 1.42 oil Objective. Images were captured in solution away from the surfaces of the slide glass and the coverslip.

siRNA and transfection

Two sets of siRNA duplexes against mouse *Matr3* were synthesized by Fasmac. The sense sequences of the siRNAs were as follows: siMatr3-1 (5′- GGGAAAAUAAAGAAUUAUA-3′), and siMatr3-2 (5′- GGAUGAUUGGGAAGAAAAA-3′). Similarly, two sets of siRNA duplexes against human *MATR3* were synthesized. Their sense sequences were as follows: siMATR3-1 (5′-CCAGAAGUAUAAAA GAAUA-3′), and siMATR3-2 (5′- GUUGGAAAAUGGAAUUAAA-3′). N2A and HEK293 cells were transfected with siRNA using Lipofectamine RNAiMAX (Thermo Fisher Scientific) according to the manufacturer's instructions. We purchased the AllStar Negative Control siRNA (1027281) from Qiagen. The efficiency and specificity of knockdown of the target genes by these siRNAs are indicated in Figures S5H and S6A.

Cell lysates and western blotting

To obtain whole cell lysates, cells were lysed with lysis buffer (50 mM Tris-HCl pH 7.4, 100 mM NaCl, 1 mM MgCl2, 0.1 mM $CaCl_2$, 1% NP-40, 0.5% sodium deoxycholate, and 0.1% SDS). After centrifugation at 18,000 × g for 5 min at 4°C, the supernatants were saved as ''whole cell lysates''. To obtain cytoplasmic and nuclear lysates, the cells were lysed with Buffer A (10 mM HEPES-KOH pH 7.8, 10 mM KCl, 0.1 mM EDTA, and 0.1% NP-40), and were centrifuged at 2000 × g for 1 min at 4°C. The supernatants were saved as ''cytoplasmic lysates''. Nuclear pellets were further resuspended in the lysis buffer. After centrifugation at 18,000 × g for 5 min at 4°C, the supernatants were saved as ''nuclear lysates''. The obtained lysates were mixed with 1xSDS loading buffer (50 mM Tris-HCl, pH 6.8, 2% SDS, 10% glycerol, 0.05% bromophenol blue, 5% 2-Mercaptoethanol). Boil the samples at 95°C for 5 min. After quick centrifugation, the samples were loaded on the SDS-PAGE gel for electrophoresis, and the proteins were transferred onto Immobilon polyvinylidene difluoride membranes (Millipore). The membranes were blocked with 1% BSA in Tris-buffered saline containing 0.05% Tween20 (TBST) for 1 hr, incubated for 1 hr with primary antibody in TBST, washed three times with TBST, and incubated for 1 hr with secondary antibody. After three washes in TBST, the blot was developed with the enhanced chemiluminescence system (Cytiva) according to the manufacturer's instructions.

CLIP-seq

HEK293 cells were transfected with the expression vectors for FLAG-tagged wild-type MATR3 (WT) and its deletion mutants lacking RRM1 (ΔRRM1), RRM2 (ΔRRM2), or IDR (ΔIDR). Similarly, HEK293 cells were transfected with the expression vectors for FLAG-mCherry-tagged wild-type U2AF2 (U2_WT) and its deletion mutant lacking IDR (U2_ΔIDR). N2A cells were treated with siMatr3-1 or siMatr3-2 to knock down *Matr3*. Additionally, we prepared two controls: N2A cells treated with control siRNA and untreated cells.

CLIP-seq was performed using the improved version of CLIP methodology, tRIP.[40] Cells were UV-irradiated at 400 mJ, and whole cell lysates or nuclear lysates (for wild-type and mutant U2AF2) were harvested from the cells as described above, except that the lysates were sonicated (MySonic, Power 30, 4 cycles of 10-s burst and 30-s rest) and treated with DNase for 10 min at 37°C prior to centrifugation. Then, tRIP libraries were generated as follows.[40] RBP-RNA complexes were immunoprecipitated with a specific antibody and Dynabeads protein G (Thermo Fisher Scientific) overnight at 4°C. After washing the beads, RNA on the beads was partially digested with RNase III (New England Biolabs) for 4 min at 37°C. Then, linker ligation was performed on beads overnight at 15°C with T4 RNA ligase 2 truncated KQ (New England Biolabs). The next day, the beads were incubated with 5′ deadenylase (New England Biolabs) for 45 min at 30°C and with Terminator exonuclease (Epicentre) for 60 min at 30°C to eliminate non-specific RNA remaining on the beads. Then, the beads were treated with proteinase K (Wako) for 40 min at 37°C and RNA fragments eluted in the supernatants were purified with Quick-RNA MicroPrep (Zymo Research). The purified RNA was reverse transcribed with SuperScript IV (Thermo Fisher Scientific) and primer digestion was performed with exonuclease I (TaKaRa) at 37°C for 30 min. Then, polyA-tailing was performed with terminal transferase (New England Biolabs) at 37°C for 50 sec, and the second-strand synthesis was performed with MightyAmp Buffer v2 (TaKaRa) at 98°C for 130 sec, 40°C for 1 min, and 68°C for 1 min. Finally, cDNAs were PCR amplified with MightyAmp Buffer v2 (TaKaRa) using the following program: 2 cycles of 98°C for 10 sec, 52°C for 1 min, and 68°C for 1 min, and then 8-20 cycles of 98°C for 10 sec, 65°C for 15 sec, and 68°C for 30 sec.

The constructed cDNA libraries were quantified with the Agilent Bioanalyzer 2100 or TapeStation 4150 (Agilent) and with QuBit high-sensitivity dsDNA assay (Thermo Fisher Scientific). The libraries were sequenced on the Illumina HiseqX with 150 bp paired-end read (Macrogen, Japan) or Miseq with 150 bp single-read (the core facility of Nagoya University). Sequenced reads were trimmed, qualified, and mapped against the mouse genome (GRCm38/mm10) or the human genome (GRCh37/hg19) as follows.[40] The obtained reads were adapter-trimmed and reads less than 18 bp were discarded using cutadapt (version 3.7).[51] Mapping was first performed against the human or mouse repetitive elements with STAR (version 2.7.9a).[52] Repeat-mapped reads were segregated for separate analysis, and all others were then mapped against the human genome or the mouse genome with STAR (version 2.7.9a). Multiply mapped reads and duplicates of reads uniquely mapped to the human or mouse genome were filtered out. For paired-end read data, only P5 reads were used for analysis. Bam files of replicated CLIP experiments were merged with samtools[63] and were used in downstream analysis, otherwise indicated. The mapping metrics are shown in Table S5.

Blue light irradiation and cell sorting

HEK293 cells were seeded in 60 mm dishes and transfected with the expression vectors for MATR3-mCherry-SspB and iLID-EGFP-FTH1. One day after transfection, cells were repeatedly exposed to blue light (1 min on, 2 min off for 12 h) using an external blue LED (470 nm) (OptoCode) placed 10 cm below the dishes at 37°C in 5% CO_2. We repeat the experiment for a technical replicate. Additionally, we prepared two control cells in which condensation was not induced: non-irradiated cells expressing MATR3-mCherry-SspB and iLID-EGFP-FTH1 and irradiated cells expressing MATR3-mCherry-SspB and EGFP alone. The cell sorting was performed on the BD FACSMelody Cell Sorter (BD Biosciences) with BD FACSChorus Software (BD Biosciences). Cells were sorted by FSC and SSC intensities and by EGFP and mCherry intensities. The sorted cells were collected into PBS containing 0.5% bovine serum albumin.

RNA-seq

Total RNA was extracted from the indicated cells using Quick-RNA Miniprep Kit (Zymo Research) according to the manufacturer's instructions. Ribosomal RNA was depleted from the total RNA with Ribo-Zero plus rRNA depletion kit (Illumina). Then, a strand-specific sequencing library was prepared with NEBNext Ultra II Directional RNA Library Prep Kit (NEB) and was read on Illumina NovaSeq 6000 (150 bp paired-end reads) at Rhelixa, Japan. Adaptor sequences were trimmed by Cutadapt (version 3.7).[51] Reads were mapped to the human reference genome (GRCh37/hg19) using STAR version 2.7.9a.[52]

Antibodies

Anti-ACTB (sc-47778), Anti-BrdU (sc-32323), anti-CELF1 (sc-20003), anti-EWSR1 (sc-28327), anti-FUS (sc-47711), anti-hnRNP A1 (sc-32301), anti-SC35 (sc-53518), anti-SF3B1 (sc-514655), anti-SNRPA1 (sc-393804), anti-SON (sc-398508), anti-U2AF2 (sc-53942) antibodies, as well as control mouse IgG, were purchased from Santa Cruz Biotechnology. Anti-PTBP1 (32-4800) and anti-U1A (PA5-27474) antibodies were purchased from Thermo Fisher Scientific. Anti-U1C antibody (70-400) was purchased from Bio Academia. Anti-U1-70K antibody (203 011) was purchased from Synaptic Systems. Anti-MATR3 antibody (ab151714) was purchased from Abcam. Anti -FUS antibody (A300-302A) and anti-TDP43 antibody (A303-223A) were purchased from Bethyl Laboratories. Anti-FLAG (M2) (P2983), anti-m3G/TMG (5′ cap) (MABE302), anti-NeuN (MAB377), and anti-pSer2 RNAP II (04-1571) antibodies were purchased from Sigma-Aldrich. Anti-PTBP1 antibody (12582-1-AP) was purchased from Proteintech. Anti-GFP antibody (11814460001) was purchased from Roche.

QUANTIFICATION AND STATISTICAL ANALYSIS

Image analysis

The images acquired with the microscopes were analyzed using Fiji (2.13.1) software.[64] The images derived from the *in vitro* droplet formation assay were quantified as follows.[71] The images were split into individual channels and the intensities of green and red fluorescence were individually adjusted between 0-65,535 arbitrary units using the Enhance contrast function (saturated = 0.35) of ImageJ. Then, the green and red fluorescence images were converted to grayscale images and combined. Masks of condensates with high fluorescence intensities were created using the Particle analysis function with thresholds of intensity = 55,000 and size = 0.2-to-infinity. The masks were applied to the original fluorescence images to measure the green and red fluorescence intensities of the condensates. For the images of cells and tissues, high-intensity regions, such as fibers and spots, were automatically identified using the custom macro programs. Briefly, nuclear masks were created using images stained with Hoechst 33258 by applying the automatic Threshold function of Fiji. To estimate the overlap of two meshworks (mesh1 and mesh2), the images were split into individual channels and converted to grayscale images after isolating the nuclear regions using the nuclear mask. Following brightness adjustment, the image of the mesh1 channel and that of the mesh2 channel were combined, and the areas occupied by at least one of the two meshworks were set as ROI (region of interest) with the automatic Threshold function (Otsu method). Then, the ROI was applied to the images of the mesh1 channel and mesh2 channel individually, and the overlap between the two meshworks was quantified using the Coloc2 function of Fiji. To analyze the circularity of mesh spots (Figure 6G), the mesh spots were detected using Fiji's automatic Local Threshold function and were set as ROI. Then, the ROI was divided into each spot with the Analyze Particles function of Fiji, and the circularity of each spot was estimated. To analyze the distance between meshwork fibers, the branch lengths, and the distribution densities of junctions and branches (Figures 1D, S1G, S1J, S6D, S6E, and S6G), images of RBP meshworks were segmented using the machine learning pixel classification tool, Labkit (Figure S1I).[65] Briefly, Labkit was trained using representative immunostaining images of RBP meshworks to create the RBP meshwork classifier against backgrounds. The classifier was consistently applied for subsequent RBP meshwork analysis. After segmentation by Labkit, the region of an RBP meshwork in an immunostaining image was further divided into individual mesh fiber regions with the Analyze Particles function of Fiji, and the number of fiber regions was counted. The distances from the centroid of a fiber region to those of all the other fiber regions were calculated using the Graph plug-in, and the minimum distance was defined as the distance from the fiber to the adjacent one. The fiber regions were further skeletonized using the Skeleton 2D/3D plugin. The numbers of branches and junctions and the lengths of branches were estimated using the Analyze Skeleton 2D/3D function of Fiji. More than twenty Z-stacked visual fields of at least four randomly selected nuclei of at least two independent experiments were analyzed in each study.

Classification of non-domain regions in the human proteome

The human protein-coding transcripts were extracted from the Ensembl reference file (Homo_sapiens.GRCh38.101.chr.gtf).[72] The transcript whose coding sequences (CDSs) were the longest and whose exons were the most numerous was selected as the representative of a gene by using original Perl and Bash scripts. The 16,424 CDSs were extracted from the representative transcripts, and their amino acid sequences were used for the downstream analysis. First, the regions annotated as domains in either CDD, Pfam, Smart, or COG database were detected using the reCOGnizer tool with default settings[32] and excluded from the analysis. The remaining regions (non-domain regions) were then divided into 80 amino acid (aa) bins, and the regions with less than 80 aa were excluded from the analysis. Disorder tendencies, hydrophobicities, and charges of amino acids were calculated using IUPred3[57] with ''long'' option, the Kyte–Doolittle scale,[73] and the EMBOSS charge algorithm (https://www.bioinformatics.nl/cgi-bin/emboss/help/charge), respectively. The mean and SD of these variables within each bin were used for the subsequent clustering analysis. The 40,133 binned regions in the human proteome were classified into 6 clusters (Figure S3B) based on their aa composition, hydrophobicity, charge, and disorder tendency using the K-means clustering algorithm with the JMP software (ver 16.1.0). Similarly, the 2,040 bins in the 521 genes for RBPs, which were associated with the GO terms; mRNA processing (GO:0006397), mRNA splicing (GO:0006380), or RNA binding (GO:0003723), were classified into 5 clusters (Figure 3B). The principal component analysis was performed using the JMP software (Figure S3A).

In Figure 3D, the enrichments of RNA-binding domains in the RBPs whose 80-aa bins belong to the five clusters were estimated using the Simple Modular Architecture Research Tool (SMART)[74] in the DAVID Bioinformatics Resources.[58,59] To analyze the localization preferences of the clustered proteins to the intracellular speckles, the lists of the proteins concentrated in the stress granules, para speckles, nuclear speckles, nucleoli, and Xist were obtained from the proteome analyses in the previous reports.[26–29] *P*-values were calculated with Fisher's exact test using R software.

In Figure 3E, the genes in the core spliceosomal components were collated by compiling genes in spliceosomal A, B, Bact, C, and P complexes, as well as in U1 snRNP, U2 snRNP, NineTeen complex, tri-snRP complex, EJC complex, SF3b complex, SF3a complex, U4/U6 snRNP, U5 snRNP, LSm proteins, Sm spliceosomal proteins, and U11 snRNP, according to the HUGO Gene Nomenclature Committee (HGNC).[35] The genes in accessory RBPs that regulate mRNA processing were obtained from the previous report[34] by filtering the genes to ''animal'' and excluding genes in the spliceosomal components. *P*-values were calculated with Fisher's exact test using R software (4.2.2).

Network and pathway analyses

In Figure S3E, the network analysis of the RBPs in each cluster was performed using the local network cluster analysis in the STRING database.[60]

In Figure 6E, the pathway analysis was performed using the Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway database[75] in the DAVID Bioinformatics Resources.[58,59] The pathways containing more than 15 genes are listed in the figure.

Motif analysis

In Figure S3D, the amino acid motifs enriched in the clusters were generated using the MEME SUITE with the following parameters ''-protein -minsites 40 -nmotifs 5 -mod anr''.[56]

Motifs enriched in the CLIP-read clusters were generated using the HOMER (v4.11)[61] with default parameters, as described elsewhere,[76] and are listed in Table S5.

CLIP-seq analysis

We used MACS2 (v2.1.4)[39] with the following parameters ''-f BAM –nomodel –broad'' to identify CLIP-read clusters. CLIP-read distributions around all coding genes, coding exons, 3′ splice sites, deep introns, and the identified peaks were calculated using bamCoverage and ComputeMatrix commands with reads per genome coverage (RPGC) normalizing from deepTools tool set (version 3.5.1).[62] The read coverages for contiguous 10 kb bins of the whole genome (Figures 5C and S5D) were calculated using multiBamSummary command of deepTools.

RNA-seq analysis

Gene expression levels were estimated using RSEM 1.3.3,[53] and differentially expressed genes were detected using DeSEQ2.[54] Percent-splice-ins (PSIs) of all internal exons and differentially spliced events were analyzed by PSI-SIGMA.[55] The comparisons were conducted between the irradiated cells expressing iLID-EGFP-FTH1 and MATR3-mCherry-SspB (two technically independent samples) and controls (the non-irradiated cells expressing iLID-EGFP-FTH1 and MATR3-mCherry-SspB, and the irradiated cells expressing MATR3-mCherry-SspB and EGFP alone).

Statistical analysis

The statistical analyses in the graphs were performed with Steel-Dwass non-parametric multiple comparisons using the JMP software (ver 16.1.0) unless otherwise indicated.